Beyond the Trait:
Present Bias as a Reaction Function

Revealed Preference, Empirical Evidence, and State-Dependent Self-Control

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Abstract

Present bias is often estimated, reported and calibrated as a personal parameter: a common β across states, goods, and settings. Re-examining data from three independent experimental designs, I test the invariance restrictions imposed by a single β and find that several fail. *First*, in a real-effort panel (Augenblick and Rabin, 2019), randomizing whether choices precede or follow a mandatory work block shifts the estimated β from 0.88 (before) to 0.74 (after) within subjects. This exogenous state variation reduces immediate effort while leaving choices and forecasts about future work unchanged, rejecting the cross-vantage equality assumed by pooled specifications. These findings are robust to design-recentering on the treatment's conditional expectation under exact randomization inference. *Second*, applying a bivariate measurement-error model to the data of Augenblick et al. (2015) to account for sampling noise reveals genuine dispersion in present bias over effort, but none over money, among the same individuals. This contradicts the stable-trait hypothesis, since no correlation can arise between a variable and a constant. *Third*, using a common protocol across rewards (Cheung et al., 2022), I confirm their finding of stronger present bias over money than over food. However, differences in β across snack subgroups exceed the money–food gap, challenging commodity-level explanations. Individual-level correlations suggest trait-like stability within trade-off structures, but not across them. Present bias thus appears to depend on the person–reward pairing, conditional on trade-off structure and state, rather than on either component alone. Uncovering these patterns requires an appropriate testing framework: the level statistics conventionally used to document heterogeneity in β cannot distinguish genuine state dependence from stochastic variation under a state-independent model. I derive a valid class of test statistics using recentered functions of the assigned state and provide a revealed-preference characterization of state-dependent quasi-hyperbolic discounting that extends Echenique et al. (2020), nesting exponential discounting and the fixed-trait specification as special cases. I also establish coordinate-level conditions for eliminating non-linear transformation bias and develop nonparametric tests that relax functional-form restrictions on utility and cost functions. To establish common ground across normative perspectives, I conservatively define welfare losses as the unanticipated component of the plan-versus-action wedge, showing empirically that it constitutes the majority of the observed wedge. Accordingly, commitment demand cannot serve as a reliable diagnostic of bias, as it tracks the accurate component of people's forecasts rather than their prediction errors. Finally, I present a framework of state-weighted valuation systems that formalizes present bias as a state-dependent reaction function, recovering traditional models under specific restrictions. Overall, these results broaden the scope of policy beyond individual targeting and voluntary self-correction to encompass the choice environment, recognizing that the decision context can embed the very cues that shape present bias.

1 Introduction

Why do people act against their own plans – procrastinating, saving less than intended, or making different choices about food, substances, or screens than they had planned? A standard answer in economics is *present bias* (PB): a systematic over-weighting of the present that makes today's self sabotage yesterday's plans. Formalized as quasi-hyperbolic discounting (Laibson, 1997; O'Donoghue and Rabin, 1999), PB has been estimated in a large body of empirical work, incorporated into models of optimal policy and behavioral interventions, and typically treated as a *personal parameter*: a single number, β, that characterizes an individual. This paper asks whether that object exists.

According to Lockwood (2020), estimates of β range from 0.34 to 1.03, a spread that no single study's confidence interval encompasses. While such dispersion is commonly attributed to imprecision, this paper advances a different possibility: the spread is not merely noise around a common value, but a trace of genuine heterogeneity in the object being measured. Before asking whether $\beta < 1$, I therefore ask a more fundamental question: does a single β exist in the first place, and what exactly is it a property of – a person, a domain, a moment, or something else?

In this sense, the analysis develops a set of ideas that have been partly present in the theoretical literature but have not been fully carried through to empirical measurement. Specifically, Loewenstein (1996) argued that the weight placed on the present increases with "visceral states," and subsequent work developed cue-triggered models in which PB varies with the salience of such cues (Laibson, 2001; Bernheim and Rangel, 2004). Pennesi (2020) derived from axioms a quasi-hyperbolic representation where the PB factor is a function of cue intensity, while O'Donoghue and Rabin (2015) identify heterogeneity across domains and states among the key open questions for the literature. Yet empirical studies typically estimate β as if it were constant. Even when the literature attempts to document heterogeneity in β, the evidence usually invoked – a profile of level estimates across groups, goods, or conditions – is methodologically unsuited to distinguish it from a fixed trait. This paper quantifies the inferential cost of assuming a constant β and provides an econometric framework to identify state-contingent PB.

Main findings. A single (β, δ) per person imposes a bundle of invariance restrictions: for instance, the same β must govern discounting for a dated payment and dated effort, for cash and a cigarette, for a rested and an exhausted person, and between an advance plan and a proximal choice. Each of the three independent designs rejects a distinct invariance restriction (Figure 1).

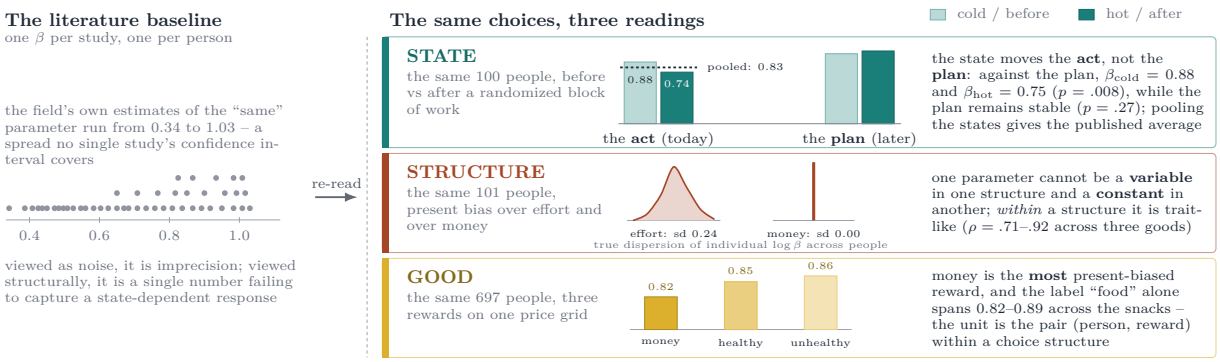
The state at the moment of choice. Augenblick and Rabin (2019) elicit real-effort allocations from 100 subjects over seven weeks. Each session requires ten mandatory "unpleasant" transcription tasks, and the authors *randomize* whether work preferences are stated before or after this first block to test for the presence of projection bias. That randomization induces within-person variation in *visceral state* at the moment of choice, crossed with choices over dated effort–wage allocations. Under this interpretation, conditioning on the state at decision time reveals that the same 100 subjects exhibit $\beta = 0.88$ before the initial set of tasks, against $\beta = 0.74$ after it ($\Delta \log \beta = -0.180$, $p = .008$). The published estimate for this panel ($\beta = 0.83$) is therefore a state-weighted average. Notably, the state manipulation has a sharply asymmetric effect: under the original paper's cost-function specification, it shifts choices about *immediate* work by -0.14 log points ($p < .01$), while leaving choices about *future* work unchanged ($+0.01$, $p = .64$). However, the projection-bias model posited by the authors applies a single uniform shift to every decision made after the mandatory work block. In other words, it assumes that visceral fatigue distorts an advance plan for a future date just as much as an immediate choice for today – imposing cross-vantage equality as an identifying restriction. The data reject this restriction ($p < .007$), revealing a distinction that the original

specification does not capture. Finally, because the experiment randomizes the *order* of a session's components rather than state labels directly, decisions are unequally exposed to the draw, given the fixed session structure. The visceral state is thus a function of an exogenous shock and of predetermined exposure, in the sense of Borusyak and Hull (2023). Recentering the treatment on its conditional expectation purges this imbalance while leaving the estimated effects virtually unchanged, yielding design-recentered PB factors of $\beta_{\text{before}} = 0.85$ and $\beta_{\text{after}} = 0.75$.

The person. Differences in average β across domains are consistent with a level shift that preserves the ranking of individuals. The relevant question is therefore whether relative differences persist, rather than whether levels coincide. I address it using a bivariate measurement-error model that separates genuine preference heterogeneity from sampling noise. In the data of Augenblick et al. (2015), this method reveals that the standard deviation of true ($\log \beta$) is 0.24 over effort and 0.00 over money, within the same people. This is fundamentally inconsistent with the stable-trait hypothesis, as a single parameter cannot be a variable in one setting and a constant in another. In contrast, when comparing choices across different rewards within the same intertemporal trade-off structure – using the three-good panel of Cheung et al. (2022) for money, healthy food, and unhealthy food – the same model recovers a high true cross-reward correlation (0.71–0.92). PB thus behaves like a trait *within* a trade-off structure, but fails to remain a single object *across* structures.

The good. A widely cited finding in this field is that the same individuals exhibit PB over effort but not over monetary rewards (Augenblick et al., 2015). This is conventionally interpreted as a structural property of commodities: money is fungible, storable, and "cold", whereas effort is experienced as it is exerted and generates immediate visceral friction. However, the data collected by Cheung et al. (2022) point in a different direction. The authors hold potential confounders fixed by employing an identical protocol across three goods – money, healthy food, and unhealthy food – for 697 subjects. While the traditional commodity reading predicts money to be the *least* present-biased of the three, their own estimates show it to be the *most*, and re-estimating their data with the scale-free estimator of Section 3.2 confirms this ordering. In particular, allowing for good-specific utility curvature, PB over money ($\beta_{\text{money}} = 0.82$) is significantly stronger than over food ($\beta_{\text{healthy}} = 0.85$ and $\beta_{\text{unhealthy}} = 0.86$, with the two categories statistically indistinguishable). Crucially, my analysis reveals that a commodity ranking is the wrong theoretical object. Within each food category, subjects selected one of three specific snacks; estimating β across these six snack-choice subgroups shows that PB spans 0.82–0.89 under the single label "food" — a range wider than the entire money–food gap. PB is therefore not an immutable property of a commodity class, but a reaction function shaped by choice architecture, immediacy, and the person–reward pair.

Figure 1: Same choices, three readings: single-parameter estimation and broken invariances.



Methodological prerequisite: repairing the test. Uncovering these empirical patterns requires first repairing the econometric toolkit, as the statistic conventionally used to evaluate state dependence cannot validly test it. Under a null hypothesis in which β is drawn from an arbitrary, heterogeneous, and serially dependent process that is *independent of the state*, a test built on the *level* of the PB wedge within state cells rejects the null of exponential discounting ($\beta = 1$) with probability one. Crucially, this approach can only detect the presence of PB, not its dependence on the state: while cell-by-cell estimates provide valid tests of time consistency, a *profile* of such estimates cannot establish that β varies with the conditioning variable. Section 3 formalizes this size distortion and characterizes the class of valid test statistics. To evaluate state dependence, any valid test statistic must project the decision-level wedge g_k (the plan-versus-action gap) onto an instrumental vector z_k that is relevant to the assigned state s_k while remaining orthogonal to baseline PB levels and predetermined confounds w (including subject fixed effects and session structure) through a balance condition, $E[z_k \mid w] = 0$. In particular, when state assignment is governed by experimental structure, the unique class of weighting functions that delivers exact size against every state-independent null without ad-hoc error assumptions consists of recentered functions of the state – formalized by the formula instruments of Borusyak and Hull (2023, 2026).[1]

This result underpins the diagnostic ladder of Section 2: its rungs translate nested structural restrictions (from unconstrained state dependence down to exponential discounting) into specific balance conditions on the weighting vector, mapping each testable step to a formal hypothesis, a valid test statistic, and an empirical verdict.

Contributions. *Empirically*, the PB factor is shown not to be a personal constant across three independent designs. *Theoretically*, a revealed-preference characterization of state-dependent quasi-hyperbolic discounting is developed, extending Echenique et al. (2020) and nesting a ladder of restrictions including the trait model. *Econometrically*, this paper demonstrates that traditional level statistics fail to evaluate state dependence, and characterizes the valid class of design-recentered formula instruments. In addition, robust testing procedures are developed, evaluated via Monte Carlo simulations, and applied to empirical panels. These include an audit of the estimator's coordinate establishing the conditions to eliminate non-linear transformation bias, a bivariate measurement-error model that separates true preference heterogeneity from sampling variance in β, and distribution-free nonparametric tests that relax functional-form assumptions on utility and cost functions. *From a welfare perspective*, this study adopts a conservative criterion that grounds welfare losses strictly in the unanticipated component of the plan-versus-outcome wedge – establishing a baseline that can command broad agreement across normative standpoints. Across diverse settings, including high-stakes addiction, most of the observed wedge stems from systematic misprediction.

[1] For example, an individual setting daily screen-time goals and facing a visceral state $s_k \in \{0, 1\}$ (where $s_k = 1$ denotes late-night exhaustion and $s_k = 0$ daytime alertness) exhibits a plan-versus-action wedge g_k for each decision k. Under any state-independent null where PB is an arbitrary stochastic process ($\beta_k \perp s_k$), a test of whether cell-level estimates differ from 1 rejects asymptotically with probability one (e.g., it could find that $\hat{\beta}_{\text{day}} = 0.85 < 1$ and $\hat{\beta}_{\text{night}} = 0.65 < 1$). A comparison of the two estimates is informative only if day and night decisions are otherwise alike: when the state marks a kind of person or follows a schedule, the difference also picks up whatever else differs between the cells. To test state dependence cleanly, one must project g_k onto an instrumental weighting vector z_k that is relevant to the assigned state s_k but orthogonal to baseline PB levels and predetermined confounds via $E[z_k \mid w] = 0$. Intuitively, z_k acts as a contrast weight that cancels out baseline individual heterogeneity – isolating whether scrolling behavior fluctuates specifically in response to state shifts, rather than reflecting the individual's time-invariant tendency to over-scroll. When state assignment s_k is governed by experimental schedules w, a recentered instrument $z_k = s_k - \mu_k$ subtracts each decision's conditional exposure probability $\mu_k = E[s_k \mid w]$. This measures whether an observation is *unexpectedly* in a visceral state relative to its schedule expectation, providing a clean exogenous shock that delivers exact size against every state-independent null.

Consequently, commitment demand cannot serve as a reliable diagnostic of bias, as it tracks the accurate component of people's forecasts rather than their prediction errors. Overall, these results broaden the scope of policy beyond individual targeting and voluntary self-correction to encompass the choice architecture, recognizing that decision environments can embed the very cues that shape PB. Finally, the Multi-System Control framework provides a *behavioral micro-foundation* for PB as a state-dependent reaction function, unifying existing approaches by nesting standard models – including quasi-hyperbolic discounting and dual-self architectures – as restricted special cases.

Related literature. On measurement and market confounders, Andreoni and Sprenger (2012a) introduced convex time budgets, while Augenblick et al. (2015) incorporated real effort and identified the domain gap re-examined below. Augenblick and Rabin (2019) constructed the multi-vantage effort panel that serves as the main empirical design here. Balakrishnan et al. (2020) demonstrated that PB over money is highly sensitive to the precise timing of immediate delivery, while Dean and Sautmann (2021), Cassidy (2019) and Andreoni and Sprenger (2012b) documented how credit constraints and risk can distort monetary wedges.

On state dependence, the conceptual foundations trace back to Loewenstein (1996), Loewenstein et al. (2003), and Krügel and Uhl (2023). Theoretical models subsequently incorporated state dependence through Laibson (2001), Bernheim and Rangel (2004), Fudenberg and Levine (2006), and Pennesi (2020), whose cue-dependent factor is generalized in Section 8. Direct empirical evidence is more limited. Psychology studies manipulate visceral states and measure monetary impatience (e.g., Badger et al., 2007; Sayette et al., 2008; Kim and Zauberman, 2013; Skrynka and Vincent, 2019), but mostly focus on behavioral proxies rather than a structural β; similarly, cognitive-load studies evaluate static impulse control without a dated trade-off (Shiv and Fedorikhin, 1999; Ward and Mann, 2000). Where quasi-hyperbolic β is estimated across states, the state is typically assigned *between* subjects and exclusively over monetary rewards. Induced sadness lowers estimated β from 0.98 to 0.94 (Lerner et al., 2013), fasting from 1.00 to 0.95 (Ashton, 2015), while a prior self-control task raises it and flattens utility curvature – the opposite of what depletion models predict (Kuhn et al., 2017). Carvalho et al. (2016) find a payday effect on monetary choices that they attribute to liquidity constraints rather than PB, with no effect on real-effort decisions. Features of the choice itself, by contrast, shift β more substantially. Delivering the "immediate" reward hours rather than minutes after the session eliminates PB (Balakrishnan et al., 2020), while across 220 published convex-time-budget estimates, reward type and delivery immediacy are the primary sources of heterogeneity (Imai et al., 2021). Existing designs, however, lack a within-person structure combining assigned states with dual-vantage elicitation – a combination needed to distinguish a state-dependent β from changes in utility curvature, baseline valuation, and measurement error.

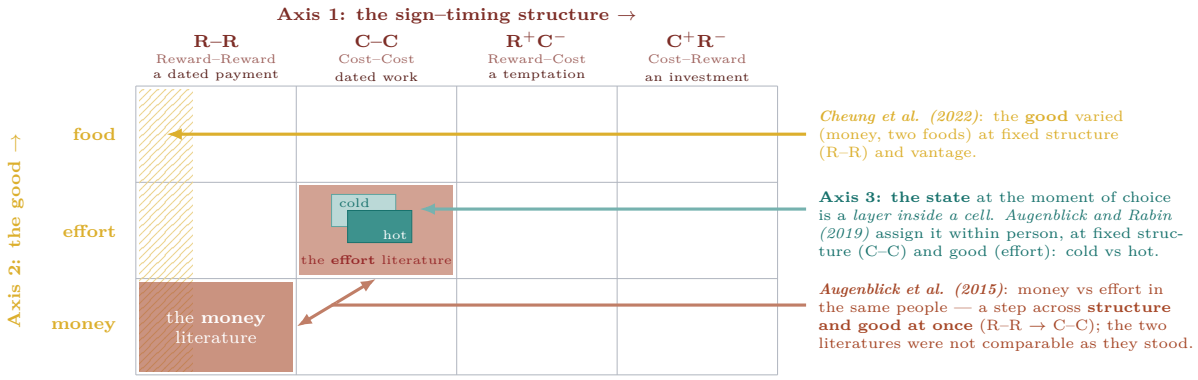
On revealed preference over time, this paper extends the axiomatic foundations of Echenique et al. (2020), building on a broader nonparametric literature on time consistency (Halevy, 2015; Browning, 1989; Crawford, 2010; Adams et al., 2014; Blow et al., 2021; Aguiar and Kashaev, 2021). For model testing and econometric inference, Adão et al. (2025) motivate the IV framework used to evaluate structural predictions, Andrews et al. (2023) and Cox and Shi (2023) provide the moment-inequality methods. Finally, Borusyak and Hull (2023, 2026) introduce the recentering approach applied here to PB.

The remainder of the paper is organized as follows. Section 2 defines the empirical object and core hypotheses; Section 3 derives the valid test statistic; Section 4 describes the data; Sections 5–7 present the empirical findings and their implications; Section 8 introduces the theoretical framework that micro-founds state dependence and reconciles standard models; and Section 9 concludes.

2 Theory: Revealed Preference and the Restriction Ladder

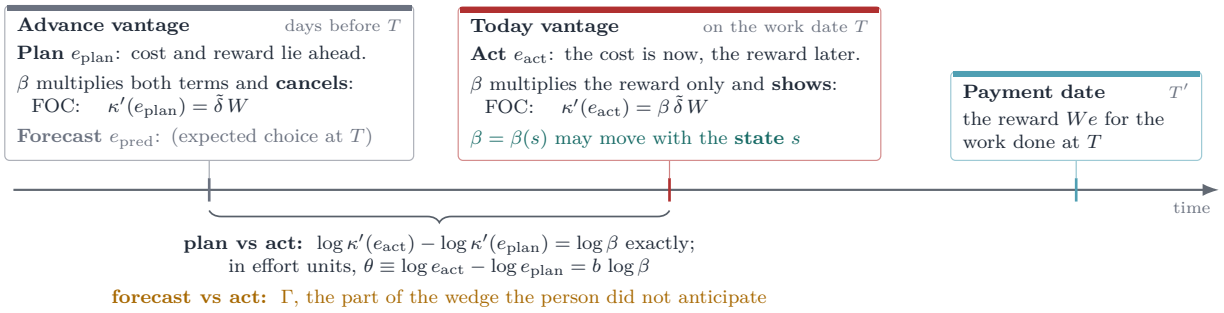
An intertemporal choice has at least three independent descriptors, and a single β must be invariant along all three: the *sign–timing structure* (Axis 1 of Figure 2), the *good* (e.g., money, effort, food, alcohol), and the *state* at the moment of choice (e.g., cued or not, hungry or sated, depleted or rested). These axes are orthogonal, but existing empirical designs typically are not. The money literature operates almost entirely in R–R structures under neutral (cold) visceral states, while the effort literature focuses on C–C structures. Consequently, comparing β_{money} and β_{effort} changes the good as well as the sign–timing structure. Each empirical design examined in this paper instead moves along specific axes while holding the others fixed.

Figure 2: The three axes and the three experiments.



A second distinction used throughout the paper is between two *vantages* (Figure 3). From sufficiently far ahead, both the cost and the reward lie in the future, so the PB factor multiplies both and cancels. The advance plan therefore reveals the long-run preference shared across selves (δ). On the day the cost is borne, instead, β multiplies the delayed reward alone, generating a plan-versus-outcome wedge. The difference between the two first-order conditions yields $\log \beta$ exactly. This identity is the starting point for every estimate in the paper (Section 3.2) and explains why the advance vantage and its prediction serve as the welfare benchmark in Section 7.

Figure 3: The two vantages: β cancels from the plan and shows in the act.



The rest of this section establishes the revealed-preference implications of state-dependent discounting, evaluating intertemporal choice consistency without parametric assumptions on utility. Section 2.1 characterizes choices rationalizable by state-dependent quasi-hyperbolic discounting, nesting the fixed-trait model and exponential discounting as special cases. Section 2.2 arranges these specifications into a restriction ladder, showing that an unconstrained $\beta(\cdot)$ carries no falsifiable implications: testability hinges on how β varies, rather than whether it does.

2.1 Revealed Preference: Characterizing State-Dependent Present Bias

Echenique et al. (2020) characterize the revealed-preference (RP) restrictions of quasi-hyperbolic discounting (QHD) with a single PB factor $\bar{\beta}$, assuming only an increasing concave utility (Afriat, 1967; Varian, 1982). I extend their characterization to a factor $\beta(c)$ that varies across exogenously assigned state cells $c \in \{1, \ldots, C\}$. In decision k, in cell $c(k)$, the agent chooses an affordable stream $x^k = (x_0^k, \ldots, x_T^k)$ at prices $p^k = (p_0^k, \ldots, p_T^k)$, where t is the delay from the decision.[2]

Definition 1 (State-dependent QHD)**.** The data are *SQH-rationalized* if there exist an increasing concave u, a $\delta \in (0, 1]$ and a function $\beta : \{1, \ldots, C\} \to \mathbb{R}_{++}$ such that every x^k maximizes $\sum_t D_{c(k)}(t)\, u(x_t)$ over its budget, with $D_c(0) = 1$ and $D_c(t) = \beta(c)\, \delta^t$ for $t > 0$. QHD is the case $\beta(c) \equiv \bar{\beta}$; exponential discounting (EDU) is $\bar{\beta} = 1$.

At an optimum, discounted marginal utility equals price multiplied by the multiplier $\lambda^k > 0$,[3]

$$D_{c(k)}(t)\, u'(x_t^k) = \lambda^k p_t^k \qquad \Longleftrightarrow \qquad u'(x_t^k) = \lambda^k p_t^k \,/\, D_{c(k)}(t), \tag{1}$$

so prices reveal marginal utilities up to three unknowns: λ^k, δ and the β of the decision's cell. By concavity, a larger quantity cannot carry a larger marginal utility. Along a sequence of n pairs of dated consumption entries, $(x_{t_i}^{k_i}, x_{t_i'}^{k_i'})_{i=1}^n$, each with a strictly larger quantity on the left than on the right, every ratio of marginal utilities is thus at most one, and so is their product:

$$1 \;\geq\; \prod_{i=1}^{n} \frac{u'(x_{t_i}^{k_i})}{u'(x_{t_i'}^{k_i'})} \;=\; \underbrace{\prod_{i=1}^{n} \frac{\lambda^{k_i}}{\lambda^{k_i'}}}_{\text{multipliers}} \cdot \underbrace{\prod_{i=1}^{n} \rho_i}_{\text{prices}} \cdot \underbrace{\delta^{-\tau}}_{\text{impatience}} \cdot \underbrace{\prod_{c=1}^{C} \beta(c)^{\,n_c^- - n_c^+}}_{\text{present bias}}, \tag{2}$$

where ρ_i is the left-to-right price ratio of pair i, $\tau \equiv \sum_i t_i - \sum_i t_i'$, and n_c^+ (n_c^-) counts the future-dated entries of cell c, which alone carry $\beta(c)$, on the left (right). The multipliers cancel if the sequence is *balanced*, that is, if each decision appears equally often on the left and the right. The impatience factor is at least one if (i) $\tau \geq 0$, since $\delta \leq 1$; and the PB factor equals one if (ii) per-cell balance holds: $n_c^+ = n_c^-$ in every cell. Then (2) reduces to $\prod_i \rho_i \leq 1$: the larger quantities must have been relatively cheaper – the *downward-sloping demand property*. The requirement that every balanced sequence satisfying (i) and (ii) exhibit this property defines the axiom **SAR-SQH**.

Proposition 1 (Characterization and nesting)**.** *The data are SQH-rationalized if and only if they satisfy* SAR-SQH*: every balanced sequence of pairs with $x_{t_i}^{k_i} > x_{t_i'}^{k_i'}$ for all i that satisfies (i) and (ii) has $\prod_i \rho_i \leq 1$. With (ii) replaced by the pooled count $\sum_c n_c^+ = \sum_c n_c^-$ this is the QHD axiom of Echenique et al. (2020); without (ii), their EDU axiom. Hence EDU $\subset$ QHD $\subset$ SQH.*

The proof (Online Appendix A) follows Echenique et al. (2020), extended to operate cell by cell.

Intuition. The test looks in the data for balanced sequences meeting (i) and (ii), on which no values of λ^k, δ or $\beta(c)$ can offset the prices in (2). A single sequence with larger quantities on the left despite a relative-price product greater than one rejects SQH: no concave u, $\delta \leq 1$ and cell-specific $\beta(\cdot)$ fit those choices. The per-cell count is a new restriction relative to Echenique et al. (2020): it excludes sequences whose bound involves a ratio $\beta(c)/\beta(c')$. For a decision k in cell 1 and k' in cell 2, the sequence $((x_1^k, x_1^{k'}), (x_0^{k'}, x_0^k))$ has $\prod_i \rho_i = (p_1^k/p_0^k)/(p_1^{k'}/p_0^{k'})$. A value above one rejects QHD, but under SQH it only implies $\beta(1)/\beta(2) \geq \prod_i \rho_i$. Data of this kind, which satisfy SAR-SQH but violate SAR-QHD, are the empirical content of state dependence.

[2] As in Echenique et al. (2020), x^k is committed to when deciding: a date is $t = 0$ on the day, $t > 0$ in advance.

[3] For exposition, u is differentiable and choices interior. A corner turns (1) into an inequality that preserves every step below; supergradients replace marginal utilities if u has kinks, which is also why ties impose no restriction.

2.2 The Ladder of Restrictions

Proposition 1 defines a ladder of restrictions of increasing stringency (Figure 4). **Rung 0** (unrestricted β) allows each state cell an unconstrained parameter $\beta(c)$. **Rung 1** (state-*independent* β) permits β to vary stochastically across periods and choices, provided its distribution is independent of the state. **Rung 2** (monotone $\beta(\cdot)$) enforces a monotonic ordering across state cells, **Rung 3** (constant β) recovers the standard trait model of QHD, and **Rung 4** ($\beta = 1$) reduces to EDU.

Figure 4: The restriction ladder, rung by rung, and the corresponding balance conditions for the test weight.

Rung	What it restricts	The weight that tests it
Rung 0 $\beta_t(X)$ free	Every cell its own β (unrestricted reaction function)	Per-cell balance (SAR-SQH(ii)): annihilates $\beta(c)$ — *nothing testable*
Rung 1 $\beta \perp$ state	The random-β null: any law for β, as long as it does not depend on the assigned state	Valid weights are the recentered functions of the assigned state (BH)
Rung 2 β monotone	β moves with the state in a direction fixed in advance	The distance-to-the-monotone-cone statistic (Wolak)
Rung 3 β constant	The trait: one β across states, structures, goods	A within-person contrast of *structures* or *goods*; a ME model
Rung 4 $\beta = 1$	Exponential discounting: no present bias at all	No balance: the level, $z = 1$ — valid *only* here

more restrictive → more testable

Each rung is a balance condition on the weight z applied to the wedge: balanced within every cell (nothing survives), balanced conditional on w (the state test), or not balanced at all (a test of existence). The level statistic answers Rung 4 and cannot answer Rung 1.

Proposition 2 (Absence of empirical restrictions on Rung 0)**.** *With the state discretized into C cells and one cell-specific wedge statistic m_c observed per cell, Rung 0 imposes no restriction on $(m_1, \ldots, m_C)$: each m_c is a strictly monotone function of β_c at fixed (u, δ), so any observed vector is exactly rationalized by some $(\beta_1, \ldots, \beta_C)$, leaving zero degrees of freedom for empirical testing.*

One might be tempted to treat *state-dependent PB* as the main hypothesis and a *fixed trait* as the alternative. Proposition 2 establishes that unconstrained state dependence is not a testable hypothesis at all: empirical content resides entirely in the structural restrictions placed on how β varies across states. Thus, evaluating state dependence requires specifying which structural restriction on parameter variation – such as those in the ladder – is actually being tested.

3 Econometric Framework for Testing State Dependence

Section 2 characterizes state dependence for decisions measured without error. When choices are observed with noise, identifying state dependence requires specifying appropriate test statistics. This section shows that the validity of such tests is governed by the same counting condition that separates SAR-SQH from SAR-QHD, derives the class of valid test statistics, shows why the conventional level statistic cannot answer a state question, defines the PB wedge underlying every empirical test in the paper, and sets out the measurement-error model used for individual factors.

3.1 Valid Test Statistics and Recentering

From an empirical perspective, the condition distinguishing SAR-SQH from SAR-QHD reduces to a statement about *weights*. A balanced sequence assigns a signed weight to each future-dated entry, $+1$ on the left and -1 on the right, and condition (ii) requires these weights to sum to zero within each state cell: $n_c^+ - n_c^- = 0$. The same logic applies to test statistics. For a PB wedge g_k of observation k (the log-wedge of Section 3.2 or an RP violation index), the test statistics considered

here take the form of a weighted average $T = \frac{1}{N}\sum_{k=1}^{N} z_k g_k$. When g_k is specified as the log-wedge, it decomposes as $g_k = b \log \beta(s_k) + \eta_k$, where $b > 0$ is the wage elasticity of planned effort defined in Section 3.2, $s_k \in \{1, \ldots, C\}$ denotes the state cell of observation k, and η_k is an error term that may depend on predetermined covariates w (such as the subject, vantage point, or wage grid).

Under any Rung-1 null, the distribution of β does not depend on the state, so the cell-specific levels collapse to a common baseline $\bar{g} \equiv b \log \bar{\beta}$, with $\log \bar{\beta} \equiv \mathbb{E}[\log \beta_k]$, and each draw's deviation from it is absorbed in the error term. This yields $g_k = \bar{g} + \eta_k$ with $\mathbb{E}[\eta_k \mid s_k, w] = \mathbb{E}[\eta_k \mid w]$: under this null, the wedge consists of a common PB level $\bar{g}$ plus an error whose conditional expectation, given predetermined variables, does not depend on the state.[4] Whether T is a valid test of this null depends on how z is balanced across the cells, and the weight can satisfy three balance conditions.

Per-cell balance, $\sum_{k:\, s_k = c} z_k = 0$ for every c, is condition (ii) applied to the weight. It removes any cell-specific level, so that T is the same whatever the value of β in each cell: a statistic balanced within cells cannot detect state dependence (Proposition 2).[5] *No balance*, $z_k \equiv 1$, yields the level statistic. Its expectation under any Rung-1 null is the mean wedge, $\bar{g} + \mathbb{E}[\eta_k]$, which differs from zero whenever PB is present and the error has mean zero: the level tests the existence of PB, not its dependence on the state. *Conditional balance*, $\mathbb{E}[z_k \mid w] = 0$, centers the weight on the assignment process, while allowing it to differ across cells. It removes the common baseline level $\bar{g}$ and every component that depends on w alone (e.g., a transformation bias, a fixed misprediction, a censoring artifact, the subject's mean wedge), ensuring $\mathbb{E}[T] = 0$ under any Rung-1 null while preserving genuine state variation in $\beta(s_k)$. A statistic with conditional balance is therefore a valid test of Rung 1.[6] Conditional balance is *recentering* in the sense of Borusyak and Hull (2023). The assigned state can be viewed as a formula instrument, $s_k = f_k(\xi; w)$: a function of an exogenous draw ξ (in Section 5, the randomized order of a session's components) and of the predetermined structure w. Its expectation over the assignment process, $\mu_k = \mathbb{E}[f_k(\xi; w) \mid w]$, can be computed, and the recentered weight $z_k = f_k(\xi; w) - \mu_k$ satisfies conditional balance by construction.[7]

Proposition 3 (The valid class for Rung 1)**.** *Under exogenous assignment, $\xi \perp \eta \mid w$:*
(i) every recentered weight $z_k = h_k(\xi; w) - \mathbb{E}[h_k(\xi; w) \mid w]$ gives $\mathbb{E}[T] = 0$ under every Rung-1 null, for any function h_k; (ii) conversely, for any weight that is a function of (ξ, w) but is not recentered, some Rung-1 null yields $\mathbb{E}[T] \neq 0$, unless further restrictions are placed on η; (iii) the level weight $z_k \equiv 1$ gives $\mathbb{E}[T] = \bar{g} + \mathbb{E}[\eta_k]$, non-zero under every Rung-1 null with PB and a mean-zero error.

Part (i) follows by the law of iterated expectations given $\mathbb{E}[z_k \mid w] = 0$, part (ii) establishes the necessity of recentering, and part (iii) illustrates the failure of uncentered level statistics ($z_k \equiv 1$), which confound the presence of PB with its dependence on the state (Online Appendix A). As a consequence, every state test used below belongs to the valid class: the recentered product moment $\mathbb{E}[(s_k - \mu_k) g_k]$, where $(s_k - \mu_k)$ measures the extent to which an observation experiences

[4] Rung 1 requires the *distribution* of β, given w, to be independent of the state; with an assigned state that leaves the other components of the error unaffected, this implies the mean condition. A weight that is a function of the state alone needs nothing more; a weight that depends on the whole assignment draw uses the independence of Proposition 3. In either case the error may vary freely with w, allowing subject-specific levels.

[5] Specifically, $\mathbb{E}[T] = \mathbb{E}\left[(1/N)\sum_k z_k g_k\right] = (1/N)\mathbb{E}\left[\sum_c \left(b \log \beta(c) \sum_{k \in c} z_k\right)\right] + (1/N)\mathbb{E}\left[\sum_k z_k \eta_k\right]$. Per-cell balance sets $\sum_{k \in c} z_k = 0$ for each c, which zeroes the first term whatever the values of $\beta(c)$. The second term does not involve β, so T has the same distribution under every hypothesis on Rung 0.

[6] Specifically, under any Rung-1 null where $g_k = \bar{g} + \eta_k$ with $\mathbb{E}[\eta_k \mid s_k, w] = \mathbb{E}[\eta_k \mid w]$, by law of iterated expectations $\mathbb{E}[T] = (1/N)\sum_k \mathbb{E}\big[\mathbb{E}[z_k g_k \mid s_k, w]\big] = (1/N)\sum_k \mathbb{E}\big[\mathbb{E}[z_k \mid w](\bar{g} + \mathbb{E}[\eta_k \mid w])\big] = 0$. Under the alternative, conditional balance does not force $\sum_{k \in c} z_k = 0$ within cells, so the cell levels survive: with a binary state and the recentered weight $z_k = s_k - \mu_k$ defined below, the same steps give $\mathbb{E}[T] = b\,[\log \beta(1) - \log \beta(0)]\, \frac{1}{N}\sum_k \mathbb{E}[\mathrm{Var}(s_k \mid w)]$, the state effect scaled by the variance of the state that remains once w is fixed.

[7] Recentering isolates the state variation remaining once w is fixed, a share $\rho = \mathbb{E}[\mathrm{Var}(s \mid w)] / \mathrm{Var}(s)$ of its variance.

an unexpected state relative to the experimental schedule.[8] Specifically, when assignment has predetermined structure w, the appropriate center is the expected state given that structure, $\mu_k \equiv \mathbb{E}[s_k \mid w]$, computed as detailed in Section 5.2.[9] Finally, the efficient member of the class is the recentered best predictor of the treatment, reweighted by the inverse conditional variance of the error (Borusyak and Hull, 2026). Figure 4 summarizes the balance condition that each rung requires. The paper takes the rungs to data in turn: Rung 1 in Section 5, where state independence is rejected for the assigned state as well as for the stake and proximity margins; Rung 2 in Section 5.4; Rung 3, the trait model, in Section 6, where it is rejected; and Rung 4 in Section 7.

3.2 Transformation-Robust Measurement: The Log-Wedge Coordinate

Estimates below are computed on a single coordinate (θ). Let e denote the effort chosen, $\kappa(e)$ an increasing, strictly convex cost, W the wage paid on a later date, and $\tilde{\delta}$ the long-run discount factor between the work date and the payment date. At the advance vantage both the cost and the reward lie in the future, so the objective is proportional to $\beta\,[-\kappa(e) + \tilde{\delta}\,We]$. The resulting first-order condition, $\kappa'(e_{\text{plan}}) = \tilde{\delta}W$, in log-linear form becomes $\log\kappa'(e_{\text{plan}}) = \log\tilde{\delta} + \log W$, which does not involve β. On the day of work the cost is immediate, the objective becomes $-\kappa(e) + \beta\tilde{\delta}\,We$, and $\log\kappa'(e_{\text{act}}) = \log\beta + \log\tilde{\delta} + \log W$. The difference between the two conditions is exactly $\log\beta$. Since effort rather than marginal cost is observed, I use the log-wedge $\theta \equiv \log e_{\text{act}} - \log e_{\text{plan}}$ and the wage elasticity of planned effort $b \equiv \partial\log e_{\text{plan}}/\partial\log W$, which the advance-vantage FOC condition identifies as the reciprocal of the elasticity of marginal cost.[10]

Then $\log\beta = \theta/b$ (exactly under an isoelastic cost, to a first-order approximation otherwise) and every reported β is obtained as $\exp(\hat{\theta}/\hat{b})$ from the within-subject regression[11]

$$\log e_{ik} = a_i + b\log W_{ik} + \theta\,\mathbf{1}\{\text{today}\}_{ik} + \varepsilon_{ik},$$

where a_i denotes subject fixed effects and ε_{ik} is an idiosyncratic error term. With $\mathbf{1}\{\text{today}\} = 1$ for choices made at the today vantage and 0 at the advance vantage, θ captures the average log-wedge between plan and action. Whenever state dependence is tested, the state-cell indicators $\mathbf{1}\{s_k = c\}$ enter both on their own and interacted with $\mathbf{1}\{\text{today}\}$, so that θ_c varies across cells while the plan itself may also shift with the state. In this coordinate, the log-wedge defined in Section 3.1 is decision-specific: for a today-decision k, $g_k = \log e_k - (\hat{a}_i + \hat{b}\log W_k)$ measures the log-deviation of the observed action from the planned allocation for the same subject at the same wage (and, when testing state dependence, within the same state). The cell parameter θ_c is the sample mean of g_k across today-decisions in cell c, and the test of state dependence evaluates the contrast $\theta_c - \theta_{c'}$: a weighted average of g_k with opposite-signed weights across the two cells, delivered directly by the interaction coefficient in the panel regression.[12]

[8] Monte Carlo simulations confirm that uncentered level weights severely overreject under state-independent nulls, whereas the recentered product moment holds size near .05 while rejecting in every replication against a genuine reaction function. More details in Online Appendix G.1.

[9] Each such moment is computed as a regression coefficient: since an OLS or IV coefficient is a weighted sum of the outcome (Frisch and Waugh, 1933; Lovell, 1963) with the residualized regressor (or instrument) as the weight (Angrist and Pischke, 2009), the choice of fixed effects, controls and instruments in Section 3.2 is the choice of z.

[10] In the advance FOC, $\kappa'(e_{\text{plan}}) = \tilde{\delta}W$ yields $\log W = -\log\tilde{\delta} + \log\kappa'(e_{\text{plan}})$. Differentiating both sides wrt $\log W$ gives $1 = \frac{\partial\log\kappa'(e_{\text{plan}})}{\partial\log e_{\text{plan}}}\frac{\partial\log e_{\text{plan}}}{\partial\log W} = \epsilon_{\kappa'}\cdot b$, where $\epsilon_{\kappa'} \equiv e\kappa''(e)/\kappa'(e)$ is the marginal cost elasticity. Thus, $b = 1/\epsilon_{\kappa'}$.

[11] Since $\log\beta = \log\kappa'(e_{\text{act}}) - \log\kappa'(e_{\text{plan}})$, an isoelastic cost $\kappa(e) = e^{\gamma}/\gamma$ with $\gamma > 1$ gives $(\gamma-1)(\log e_{\text{act}} - \log e_{\text{plan}}) = \log\beta$, which implies $\theta = b\log\beta$, given $b \equiv \partial\log e_{\text{plan}}/\partial\log W = 1/(\gamma-1)$. Testing state dependence does not require isoelasticity: at a given wage the plan does not involve β, and since κ' is strictly increasing the act is strictly increasing in β, so $\theta_c = \theta_{c'}$ if and only if $\beta_c = \beta_{c'}$ whenever the cells face the same wages, as they do in Section 5.

[12] Subject fixed effects recenter these weights within each respondent, achieving conditional balance at the subject

An advantage of the coordinate θ is that it is estimated directly from observed choices, avoiding functional-form assumptions on the cost structure and preserving the nonparametric nature of the test. Inference in θ is linear in the data and never divides by an estimated elasticity. Specifically, θ satisfies three distinct econometric properties. First, it is *scale-free*: the level difference $e_{\text{act}} - e_{\text{plan}}$, reported in parts of the literature, grows mechanically with the planned quantity at a fixed β, whereas the log-wedge does not. Second, at a common b, it is *curvature-free*: since $\theta = b \log \beta$ combines preference with cost curvature, θ compares β across cells whenever b is shared.[13] Third, it is *transformation-robust*: tests in θ are unaffected by the convex map to $\hat{\beta} = \exp(\hat{\theta}/\hat{b})$, whose expectation exceeds β by a factor of approximately $1 + \frac{1}{2}\text{Var}(\hat{\theta}/\hat{b})$, a term amplified by $1/b^2$. This convexity bias is negligible in aggregate (+0.07% on the effort panel) but substantial at the individual level. On the data of Augenblick et al. (2015), the mean of unrestricted per-subject effort $\hat{\beta}$'s is 1.38, against 0.92 once each estimate is corrected: an inflation of 49.9%. Imposing their common-curvature restriction ($b_i = b$) reproduces their published mean (0.916). Crucially, allowing fully heterogeneous per-subject curvature (b_i) and correcting for this non-linear transformation bias yields virtually the same result (0.92), showing that the finding is robust to relaxing common curvature once the convexity bias is accounted for. Every statistical test is therefore conducted in θ (or θ/b), and reported point estimates are converted to β solely for ease of interpretation (*two-step* procedure).

3.3 Accounting for Sampling Noise: A Bivariate Error Model

Evaluating whether PB reflects a stable personal trait requires estimating individual parameters – a setting where sampling instability poses a more severe challenge than non-linear transformation bias. Specifically, given the limited number of decisions per respondent, estimating a subject-specific elasticity $\hat{b}_i$ introduces extreme noise into the denominator.[14] Following Augenblick et al. (2015), I resolve this instability by estimating individual factors under a common domain-level elasticity $\hat{b}$ alongside subject-specific today-effects $\hat{\theta}_i$, yielding $\log \hat{\beta}_i = \hat{\theta}_i/\hat{b}$. Under this formulation, each $\log \hat{\beta}_i$ is linear in the subject's own choices, avoiding heavy-tailed ratios and allowing the remaining sampling noise to be modeled directly. For a subject observed across two domains $d \in \{1, 2\}$, the pair of estimated log present-bias factors is specified as:

$$\hat{r}_i = r_i + \nu_i, \qquad r_i \sim N(\bar{r}, \Sigma), \qquad \nu_i \sim N(0, V_i), \qquad V_i = \text{diag}(V_{i1}, V_{i2}),$$

where $r_i = (\log \beta_{i1}, \log \beta_{i2})'$ contains the subject's true preference parameters, Σ is the population variance-covariance matrix of interest, and V_{id} is the sampling variance of the estimate in domain d.[15] Assuming sampling errors independent across domains and of the true preferences r_i, the marginal distribution of observed estimates is $\hat{r}_i \sim N(\bar{r}, \Sigma + V_i)$, and $(\bar{r}, \Sigma)$ can be estimated via maximum likelihood.[16] The diagonal elements of Σ represent the true population variance of $\log \beta$ within each domain, purged of sampling noise. Isolating these diagonal entries directly addresses the

level; when state assignment follows a structured schedule (as in Section 5), the state and its interaction are instead instrumented by their recentered versions, $s_k - \mu_k$ and $\mathbf{1}\{\text{today}\}(s_k - \mu_k)$, or μ_k is included directly as a control – the two asymptotically equivalent formulations of Borusyak and Hull (2023).

[13] This holds across the state cells of Section 5 (where it is tested and not rejected), but fails across the goods of Section 6.2, where comparisons are instead made in θ/b.

[14] Evaluating the ratio $\hat{\theta}_i/\hat{b}_i$ subject by subject divides by an elasticity estimated on a small set of choices. When a subject barely responds to prices ($\hat{b}_i \approx 0$), the ratio becomes heavy-tailed: in the effort task of Augenblick et al. (2015), a per-subject bootstrap reveals that $\hat{\beta}_i$ has no usable mean for 26 of 86 subjects.

[15] Where microdata are available, V_{id} is computed from the subject's residual variance and number of choices at each vantage, divided by $\hat{b}^2$; when relying on published estimates, $V_{id} = (\text{se}_{id}/\hat{\beta}_{id})^2$, the delta-method variance of $\log \hat{\beta}_{id}$.

[16] Through the inverse-variance weighting $(\Sigma + V_i)^{-1}$, imprecisely estimated subjects receive lower weight in the likelihood function, preserving sample size while preventing noisy estimates from distorting inference.

prerequisite that any cross-domain correlation presupposes: whether genuine preference dispersion exists within a given domain in the first place. Section 6 applies this model to effort and money in Augenblick et al. (2015) and to the three goods of Cheung et al. (2022); Online Appendix D provides further details on estimation, tests, and robustness.

4 Data

Table 1 summarizes the experimental designs of the three primary datasets supporting the empirical findings below. In Augenblick and Rabin (2019), hereafter *AR2019*, 100 subjects allocate transcription tasks across seven sessions over six weeks under piece rates ranging from 1 to 35 cents, making choices for the same dated work between 0 and 26 days in advance alongside incentivized predictions (11,405 decisions in total). Each session features ten mandatory tasks, with the ordering of session components block-randomized across subjects. In Augenblick et al. (2015), hereafter *ANS2015*, the same 101 subjects complete intertemporal allocation tasks over both real effort and money within separate, parallel protocols conducted over seven weeks. In Cheung et al. (2022), hereafter *CTW2022*, 697 students complete convex time budget elicitations across three reward domains – money, healthy food, and unhealthy food – under an identical price grid, time horizon, and delivery protocol. Measured over money, PB estimates may confound the discount factor with other features of dated payments – risk and trust in the experimenter, transaction costs, and the possibility of borrowing or saving outside the experiment (e.g., Andreoni and Sprenger, 2012b; Dean and Sautmann, 2021). The three experimental setups neutralize these channels across the trade-offs analyzed in this paper. [17] Nineteen further datasets provide supplementary results in subsequent sections and are documented comprehensively in Online Appendix I.

Table 1: The three designs.

Dataset	Decision made	Cost borne	Reward	Cell	What varies
AR2019 Augenblick & Rabin (2019) 100 subjects · real effort	3–26 days before the work date, and on the day itself	the work date	one fixed date, 7 weeks after the start	C–C	**just completed 10 mandatory tasks** (randomized order); the wage
ANS2015 Augenblick, Niederle & Sprenger (2015) 101 subjects · effort, money	effort: weeks 1 and 4 (advance), weeks 2 and 5 (on the day) money: weeks 1 and 4	effort: the two work dates money: —	effort: a fixed bonus, week 7 money: two dated payments	C–C R–R	the structure and the good, in the same people
CTW2022 Cheung, Tymula & Wang (2022) 697 students · three goods	week 1 (both rewards ahead) and week 2 (sooner reward today)	—	dated money or food, weeks 2 and 5, delivered identically	R–R	the good; the vantage

[17] In AR2019 and in the effort task of ANS2015, all work is paid on one fixed date, the same for the plan and the act, so payment risk, trust, transaction costs and liquidity enter both choices alike and cancel from the wedge; effort, unlike money, also cannot be moved across dates by outside borrowing or lending. In CTW2022 the three rewards share dates, delivery and presentation, and Online Appendix E shows that arbitrage, trust and the scale of delivery do not account for the ordering of goods. Moreover, any remaining factor that does not depend on the randomized order is balanced across the assigned states, and recentering removes every component that depends on the predetermined structure alone (Section 3.1).

5 State Dependence

5.1 Assigned Decision States Move β

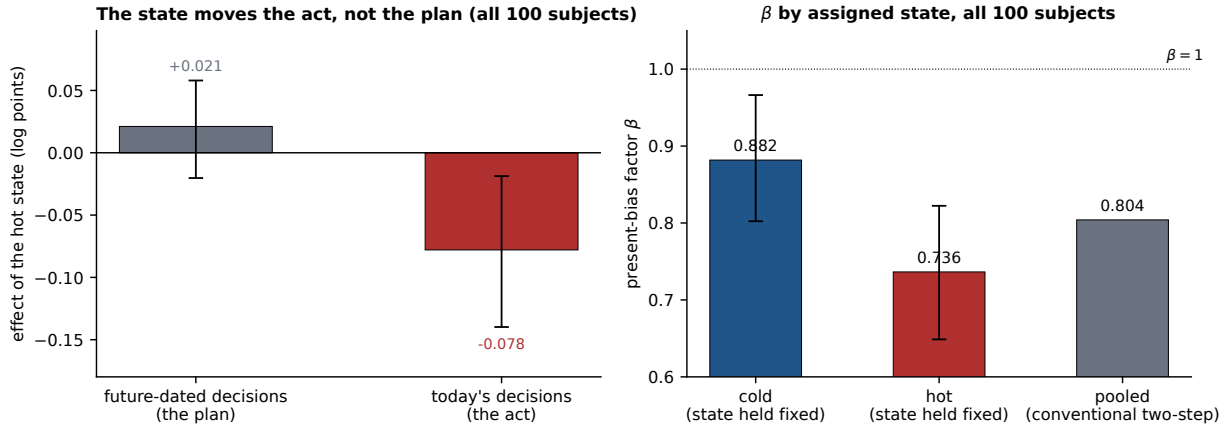
A recurring challenge in the literature on visceral influences is that a "state" is often confounded with person-level attributes. Hungry, fatigued, or intoxicated individuals may differ along many unobserved dimensions, so that comparing across states inadvertently conflates state dependence with individual heterogeneity. Section 3 sharpens the test of state dependence into two requirements: only an assigned state can reject the random-β null, and only a recentered statistic can isolate the effect of the state. The experimental design of Augenblick and Rabin (2019) allows both criteria to be met, yet had not previously been exploited for this purpose. Crossing the assigned state (deciding before versus after the mandatory task block) with the decision vantage (a future-dated plan versus a today-action) generates a 2×2 factorial design on a common wage grid within the same subjects: a plan formed in a cold state, a plan formed in a hot state, an action taken in a cold state, and an action taken in a hot state. Estimating the two-step specification with state and vantage fully interacted within subject, evaluated via a subject bootstrap, yields the following results.[18]

Table 2: Present bias is state-dependent: it intensifies in the hot condition.

	All 100 subjects		AR2019 subset ($N = 72$)	
wage elasticity b	0.549	[0.489, 0.615]	0.654	[0.602, 0.709]
hot state on a *future*-dated choice	+0.021	[−0.020, +0.058]	+0.006	[−0.039, +0.046]
β, state held *cold*	0.882	[0.802, 0.966]	0.904	[0.833, 0.981]
β, state held *hot*	0.736	[0.649, 0.822]	0.748	[0.654, 0.841]
$\Delta \log \beta$ across the assigned state	**−0.180**	[−0.310, −0.062]	−0.191	[−0.326, −0.060]
p-value	.008		$< .004$	

Note: Within-subject estimates; subject bootstrap ($B = 499$); 95% intervals. Main results use the full sample; results on the original paper's estimation subsample are shown alongside.

Figure 5: The hot state cuts immediate effort while leaving future-work decisions unchanged; pooling the states, as AR2019 do, returns a state-weighted average.



[18]Throughout, "subject bootstrap" refers to resampling whole subjects with replacement, re-estimating the full two-step procedure across B resamples, and deriving confidence intervals and two-sided p-values directly from the resulting empirical distribution; this accounts for arbitrary dependence across choices made by the same individual, within whom the state is assigned, session by session.

The pooled β for this panel, like the published one, is a state-weighted average (Figure 5): a study whose observation protocol sampled hot and cold states in different proportions would have inferred a different parameter. Because the state is exogenously assigned within subject, this finding rejects not merely a constant β, but the entire class of state-independent present-bias processes (Rung 1 Wald test of state equality: $\chi^2_1 = 9.0$, $p = .003$ on AR2019's 72-subject subsample). While AR2019 restrict their main analysis to the 72 subjects for whom structural estimation succeeds, my two-step procedure retains all 100 subjects with a common domain elasticity $\hat{b}$, leaving the core finding intact ($\Delta \log \beta = -0.180$, $p = .008$). Under minimal functional-form assumptions, the direction of this finding is corroborated: matching each today-choice directly into the advance schedule bounds β nonparametrically between wage grid points. Computed separately by state, the cold-state bracket lies above the hot-state bracket at both wage binnings, meeting only at the boundary (Online Appendix C). Finally, state assignment is orthogonal to the wage grid (16.2 cents in cold sessions and 16.3 in hot sessions), confirming that state dependence is not driven by price composition.

5.2 Unequal Exposure: Design Recentering and Randomization Inference

The experiment randomizes the *order* of a session's components, rather than assigning a state label directly to each choice. A session day comprises up to five decision sets (today's work, one or two sets of future work, and one or two sets of predictions) alongside one block of ten mandatory tasks; a decision is *hot* if its set was administered after this block. Whether a decision falls into the hot state is a deterministic function of the random order draw ξ and the day's predetermined menu structure w: a formula instrument, $s_k = f_k(\xi; w)$, characterized by non-random exposure to exogenous shocks. Exposure to the hot state varies systematically across decision types: today's work is hot 46% of the time, the first future set 60%, and the second future set 83%. To address this, the expected state $\mu_k \equiv \mathbb{E}[s_k \mid w]$ is computed as the share of subject-days within stratum w (calendar day $\times$ menu composition) in which choice k fell after the mandatory block ($s_k = 1$).[19] This expected treatment μ_k explains 19% of the variance of the raw state indicator s_k and accounts for the correlations of s_k with the decision vantage (-0.22), the distance to the work date ($+0.15$), and the decision day (-0.12).[20] Subtracting μ_k drives all three correlations to essentially zero (Figure 6), redefining the instrument as the realized (exogenous) state assignment relative to expected exposure rather than a composite of random shocks and structural baseline variation.[21]

Hence, the state effect is estimated via 2SLS using the recentered instrument (instrumenting the state indicator and its interaction with vantage by their recentered counterparts), via a control-function specification (adding μ_k and $\mu_k \times$ today as controls), and in reduced form. All three approaches, which Borusyak and Hull (2023) show to be valid under non-random exposure, yield virtually identical results.[22]

[19] An unadjusted indicator $s_k \in \{0, 1\}$ would conflate state dependence with session timing: hot choices would over-represent future dates (83%), whereas cold choices would disproportionately involve present work (54%). In addition, shifts in available time horizons across experimental weeks introduce further mechanical variation. Subtracting μ_k constructs a recentered instrument $\tilde{s}_k \equiv s_k - \mu_k$ with zero conditional expectation ($\mathbb{E}[\tilde{s}_k \mid w] = 0$), eliminating the spurious over-representation of future choices in late-session slots.

[20] These correlations do not reflect behavioral preferences, but rather an artifact of the experimental software's order-generation protocol that must be purged to isolate the exogenous variation in state assignment.

[21] Estimating μ_k relies on the assumption that realized orderings are exchangeable within each stratum – an empirically testable condition. As reported below, the recentered instrument satisfies the joint falsification test ($p = .28$), while identifying the distance to the work date as the sole covariate requiring explicit control.

[22] Under the exchangeability assumption, component orderings are equally likely within stratum w. Under the null (the state affects no decision), choices remain fixed across counterfactual orderings; p-values are obtained by redrawing orderings 999 times within strata and comparing the observed contrast against the randomization distribution.

Table 3: The state effect, unadjusted and design-recentered.

	All 100 subjects		AR2019 subset ($N = 72$)	
	unadjusted	recentered	unadjusted	recentered
hot state on the plan	$+0.021$ ($p = .27$)	-0.010 ($p = .68$)	$+0.006$ ($p = .79$)	-0.030 ($p = .20$)
hot state on the act	-0.078 ($p = .02$)	-0.077 ($p = .01$)	-0.119 ($p < .004$)	-0.136 ($p < .004$)
exact RI p	—	**.018**	—	**.001**
β cold / hot	0.882 / 0.736	0.847 / 0.750	0.904 / 0.748	0.881 / 0.750
$\Delta \log \beta$	-0.180 ($p = .008$)	-0.122 ($p = .12$)	-0.191 ($p < .004$)	-0.161 ($p = .02$)
exact RI p	—	**.091**	—	**.021**

Note: Subject-bootstrap p in parentheses ($B = 499$). Exact randomization inference (RI) over 999 redraws of the session orderings within strata. The act row is the total effect of the state on today's decisions.

Figure 6: Purging non-random exposure by recentering the instrument.

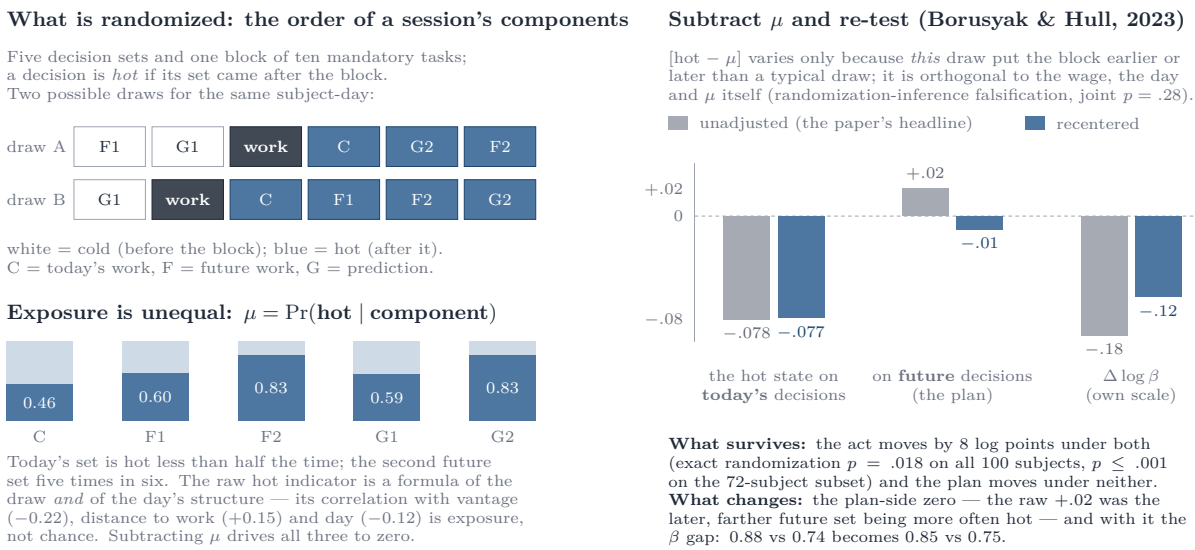
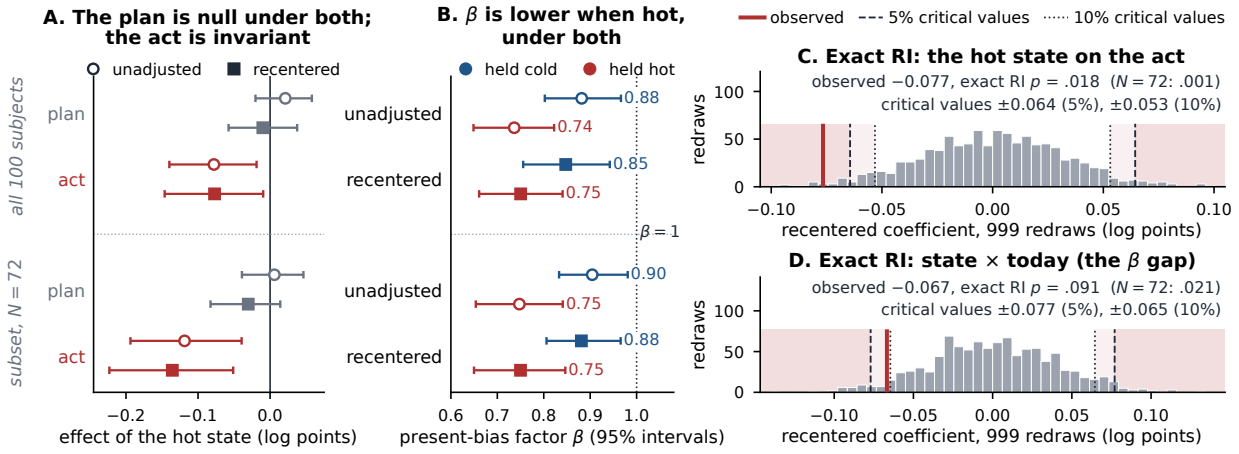


Figure 7: Recentering moves the plan-side zero, not the act: the hot state lowers today's allocation under both estimators and leaves the plan untouched, and β is lower when hot under both.



Note: The state effect, unadjusted and design-recentered. *(A):* the effect of the hot state on the plan and on the act, log points, 95% subject-bootstrap intervals. *(B):* β with the state held cold and held hot, under each estimator and in each sample, 95% subject-bootstrap intervals. *(C), (D):* the exact randomization-inference null distributions on all 100 subjects (the recentered coefficient under 999 redraws of the session orderings within strata), the critical values, and p-values.

Table 3 and Figure 7 report the results. The effect of the hot state on the act is invariant to recentering: today's allocation falls by 7.8 log points unadjusted and by 7.7 recentered, with an exact randomization p of .018 on all subjects and at most .001 on the AR2019 subset. The plan remains unaffected under both specifications (+0.021 and −0.010; randomization $p = .73$).

What recentering changes is the counterfactual against which the act is measured. The raw +0.02 on future-dated decisions reflects exposure rather than the state, since the second future set refers to a later date and is hot more than 80% of the time. Once this exposure is removed, the gap between the cold and hot β narrows from 0.15 to 0.10 (0.85 against 0.75) – driven by the adjustment in the plan estimate, which nonetheless remains a statistical zero. [23]

5.3 The State Shifts the Act but Leaves the Plan Unchanged

AR2019 randomize session order to examine projection bias, but focus on pooled effects and leave differences across decision vantages uninterpreted. Their projection-bias model posits that agents extrapolate current marginal disutility to all subsequent choices, imposing a uniform shift across post-work decisions (present work, future work, and predictions). *Cross-vantage equality* – an identifying restriction of this structural parameter – is tested and rejected in Table 4.

Table 4: The cross-vantage test, unadjusted and design-recentered.

effect of the hot state on	unadjusted	recentered
present decisions	-0.141 $[-0.222, -0.053]$	$\mathbf{-0.160}$ $[-0.254, -0.067]$
future decisions	$+0.008$ $[-0.030, +0.048]$	-0.031 $[-0.074, +0.015]$
predictions	$+0.015$ $[-0.029, +0.062]$	-0.013 $[-0.069, +0.043]$
present − future	-0.149 $[-0.246, -0.062]$, $p < .007$	$\mathbf{-0.129}$ $[-0.227, -0.035]$, $p = .007$

Note: Source paper's coordinate and estimation sample (8,049 decisions, 72 subjects); 95% intervals.

The effect AR2019 attribute to projection bias is estimated from a pooled contrast whose identifying variation is in fact driven almost entirely by immediate decisions. This matters critically for interpretation: projection in the sense of Loewenstein et al. (2003) has no scope for immediate choices,[24] and under their model the bias should be flat across time horizons, whereas in the data it falls from −0.14 for work chosen minutes ahead to about zero for work chosen days ahead. [25]

[23] The recentered state is orthogonal to the wage drawn, the net distance to payment, the decision day and the expected treatment itself (randomization $p = .58$, .53, .16 and .82; joint $p = .28$). It fails only on the distance to the work date (+0.085, $p = .011$, against +0.55 for the raw indicator), which records the future dates a day's decision sets refer to; the distance is therefore added as a control, and moves neither estimate of the act by more than 0.001. The stakes margin of Section 5.4 requires no such correction: the wage draw is flat across components, days, vantages and states, so its expected instrument is a constant.

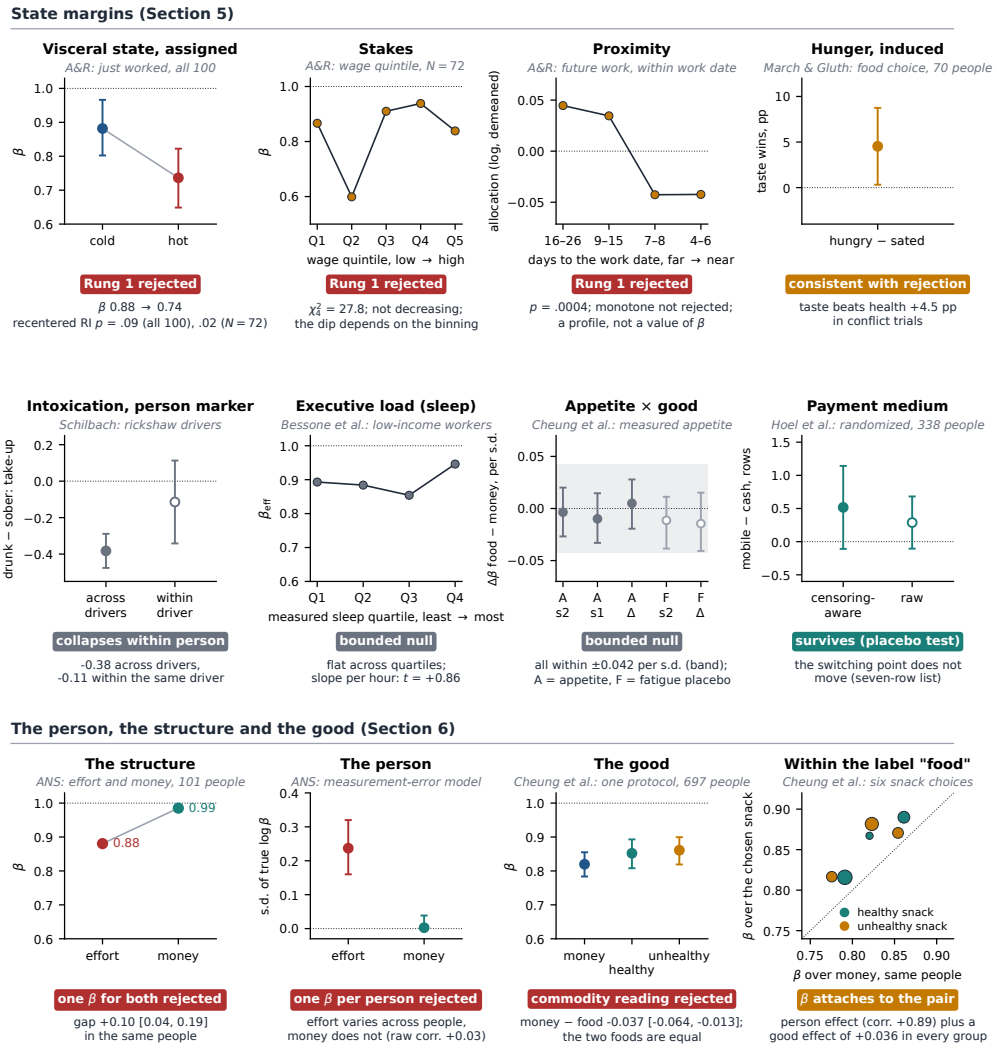
[24] Because the work is performed immediately, the state at the time of choice coincides with the state at execution.

[25] The total raw wedge ($\theta = -0.18$) decomposes as follows: the state's effect on the plan accounts for −3% $[-34, 18]$ (indistinguishable from zero), its differential effect on the act (the state × today interaction) accounts for 68% $[26, 115]$, and the remaining 36% $[11, 69]$ reflects PB with the state held fixed. The cost-curvature parameter b is virtually identical across the cold and hot states (0.657 vs. 0.654), which ensures that comparisons in effort wedges (θ) directly mirror shifts in time preferences (β).

5.4 Stakes, Proximity, and Other Margins

Applying this framework across other state margins and datasets yields the results summarized in the upper section of Figure 8. For *stakes*, flatness of β across wage quintiles is uniformly rejected ($p \leq .03$ in every specification and binning), a pattern not driven by any single subject (72 leave-one-out refits, all $p < .001$). The monotonicity restriction is also rejected at every binning, confirming that β does not decline with the stake. Evaluating the *temporal proximity* of future choices reveals that allocated effort for a given work date systematically declines as that date approaches ($p = .0004$). March and Gluth (2025) manipulate *hunger* within seventy subjects; in trials featuring a taste–health conflict, the tastier option is chosen over the healthier one in 66.7% of decisions when hungry, compared to 61.4% when sated (+4.5 pp, $p = .035$). In contrast, *cross-sectional markers* fail to persist under within-person comparison: in Schilbach (2019), intoxicated drivers take up commitment 43% of the time versus 81% when sober (-0.38, $t = -8$), yet comparing the same driver's choices when sober and when intoxicated at the interview, the effect shrinks to -0.11 ($t = -1.0$). Likewise, binning workers in Bessone et al. (2021) by measured sleep (evaluating *executive load*) reveals a flat trajectory for β_{eff} – an estimated null that bounds the effect against dual-self predictions. In the same vein, appetite variations in Cheung et al. (2022) fail to shift the food-minus-money β gap significantly. Overall, Rung 1 is rejected on margins where the state is exogenously assigned within a person, whereas it is not rejected when state variation is driven by cross-subject characteristics.

Figure 8: Where β moves and where it stops



6 Present Bias Across Individuals, Goods, and Structures

6.1 Failure of the Stable-Trait Hypothesis

Comparing time preferences across domains, Augenblick et al. (2015) elicit real-effort and monetary discounting within the same 101 subjects, finding substantial PB over effort but little over money. Re-estimating the aggregate data recovers the same disparity ($\beta_{\text{effort}} = 0.88$ versus $\beta_{\text{money}} = 0.99$). The gap is fully robust to accounting for monetary corner choices via a Tobit model and to expressing the gap in $\log \beta$, where non-linear transformation adjustments shift the estimate by only 1%. To evaluate whether β is stable across domains at the person level, I apply the measurement-error framework in Section 3.3, using domain-specific elasticities (Online Appendix D), and find:

$$\begin{aligned} \text{Effort:} \quad & \text{SD}(\text{true } \log\beta) = 0.24 \quad [0.16,\ 0.32], \\ \text{Money:} \quad & \text{SD}(\text{true } \log\beta) = 0.00 \quad [0.00,\ 0.04]. \end{aligned}$$

For effort, the plan-versus-act wedge exhibits genuine population heterogeneity beyond sampling noise,[26] whereas PB over money displays no such variation.[27] Because population PB over money is effectively constant while that over effort varies across individuals, population correlation is undefined; empirically, the sample correlation between individual point estimates is just $+0.03$ $[-0.10, +0.15]$. These results are inconsistent with PB as a stable person-level trait across domains.

Applying the measurement-error model within a given sign–timing structure using data on monetary, healthy, and unhealthy food rewards from Cheung et al. (2022) reveals a different pattern. In that dataset, both goods in each pairwise comparison exhibit genuine population dispersion (true SD $\sigma \in [0.20, 0.23]$), and high cross-reward correlations.[28] Taken together, these results suggest that β behaves as a stable trait within a given sign–timing structure, whereas it does not constitute a single invariant object across structures. This distinction is consistent with findings in the literature, including the temporal stability of monetary β (Meier and Sprenger, 2015) and the weaker stability of experimental measures in field samples (Chuang and Schechter, 2015).[29]

6.2 Commodity Categories and Person–Reward Pairings

In the literature, cross-domain disparities in PB are typically interpreted as intrinsic properties of the underlying commodities, a view that potentially conflates distinct axes of variation (Figure 2). This interpretation can be tested using the data from CTW2022, which hold fixed several confounding factors present in conventional comparisons: their design uses a single protocol across 697 subjects and two sessions, with three rewards evaluated on the same convex-time-budget grid over the same dates, delivered in identical bags and presented with matching screen representations.[30] Under the *commodity* hypothesis, money should exhibit the least PB; under the *temptation* interpretation, unhealthy food should exhibit the most PB. Applying the scale-free two-step model with good-specific elasticities yields the estimates in Table 5, consistent with the ordering reported by CTW2022.

[26] The hypothesis of zero dispersion is rejected ($p < .00001$), even under leave-one-out robustness tests.

[27] Among the half of subjects with the most precise monetary estimates, the observed dispersion is 0.059 against a median sampling standard error of 0.077, indicating that the entire spread reflects sampling noise.

[28] Specifically: money–healthy $+0.92$, money–unhealthy $+0.89$, and healthy–unhealthy $+0.71$ (with the caveat that all three are elicited in the same sittings, preventing the removal of a common-occasion component).

[29] This conclusion is consistent with Cerrone et al. (2026), who report a rank correlation of 0.77 between their individual effort and money parameters. Since both parameters are estimated from the same effort choices, the correlation is measured within one trade-off structure (effort exchanged for pay).

[30] Although the design does not randomize immediacy, all three rewards are elicited from every subject in both sessions, so common session-level shocks cancel out in the cross-reward contrast. The identified object is the differential today-effect across goods, controlling for subject $\times$ reward fixed effects.

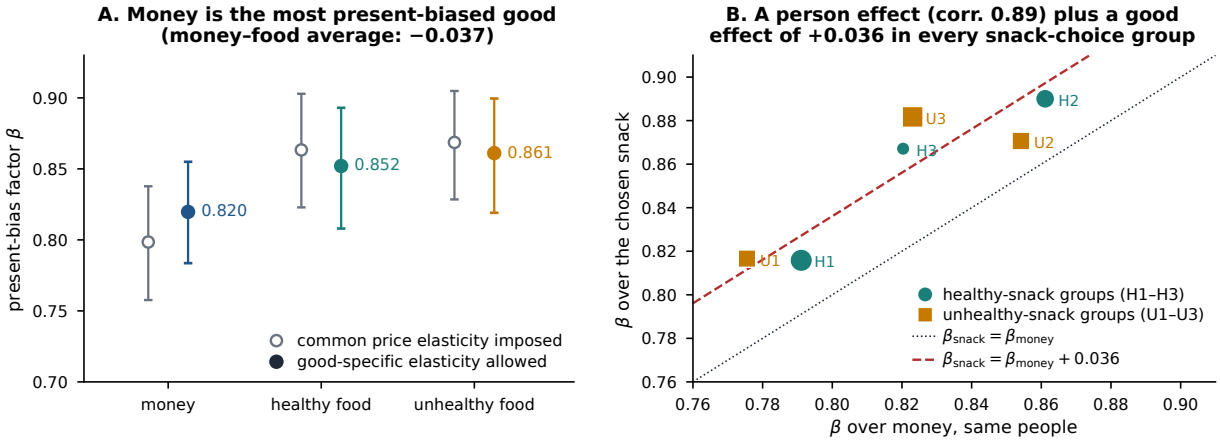
Table 5: Present bias varies across goods and is stronger for monetary rewards.

Good	elasticity b	shift θ	β
money	1.713	−0.341	**0.820** [0.784, 0.855]
healthy food	1.398	−0.224	0.852 [0.808, 0.893]
unhealthy food	1.425	−0.213	0.861 [0.819, 0.900]
money − food average		**−0.037**, $p < .004$	
money − healthy		−0.032, $p = .052$	
money − unhealthy		−0.041, $p = .004$	
unhealthy − healthy		+0.009, $p = .73$	

Note: Three goods, one protocol, within subject and within good; good-specific price elasticities; subject bootstrap, $B = 499$. With a common elasticity imposed the money-minus-food gap is −0.068; the elasticities differ by +0.32 and +0.29 ($p < .004$), and freeing them absorbs 45% of it.

These findings reject both hypotheses, as PB is strongest for money and indistinguishable between the two food types.[31] While CTW2022 suggest that monetary PB may reflect features of their non-WEIRD, low-income adolescent sample, they do not explain why it exceeds PB over food. By disaggregating choices within each food class, I find that β varies across specific snacks by more than the overall money–food gap: it ranges from 0.82 to 0.89 for healthy options and from 0.82 to 0.88 for unhealthy ones.[32] This spread can be decomposed because every subject also made choices over money. In fact, grouping subjects by their chosen snack, the cross-group standard deviation of β is 0.033 over food and 0.034 over money for the same individuals, with the two group means correlated at +0.89. This tight co-movement indicates that snack-level variation is predominantly a *person* effect, accompanied by a *good* effect of +0.036 on average that raises β for food relative to money across all six groups. Consequently, a standard commodity taxonomy fails to organize these data (Figure 9): the broad label "food" spans $\beta \in [0.82, 0.89]$, while money's own β ranges from 0.84 among non-substance users to 0.74 and 0.76 among drinkers and smokers.

Figure 9: Money is the most present-biased good, and within food β follows the person who chooses the snack.



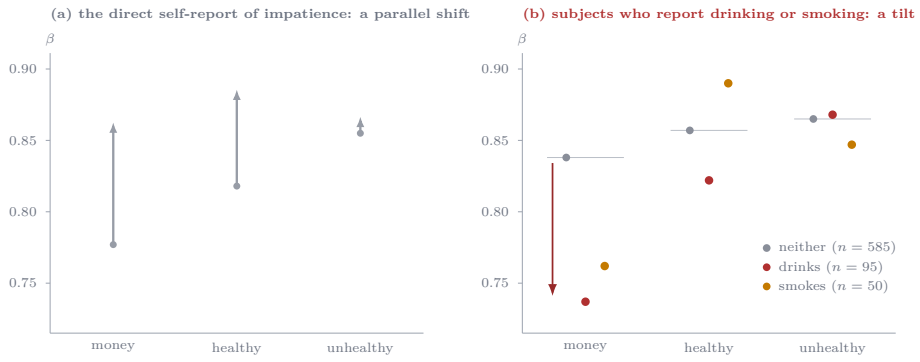
(A): β by good, 95% CI. *(B):* the 6 snack-choice groups, β over the chosen snack against β over money for the same people (marker size proportional to the group): a person effect along the diagonal, and a good effect displacing every group above it.

[31] This ordering is robust across five estimators – including an interval regression that accounts exactly for the design's six-bundle discreteness – and across both arms of the randomized presentation order. Appendix E shows that it is not driven by arbitrage, trust in the experimenters, stated valuations of the rewards, or the scale of delivery.

[32] Specifically, PB is strongest for the snacks preferred by individuals with high sweet-craving scores.

This contrast sheds light on a potential driver of the monetary PB disparity. A plausible mechanism is that, for low-income adolescents, cash provides a general claim on discretionary immediate consumption, whereas a single food item offers only a non-fungible reward tied to a specific consumption good. In particular, subjects reporting alcohol ($N = 95$) or tobacco use ($N = 50$) may face immediate cash needs that no single snack can satisfy. Crucially, their estimated β_{money} falls to 0.74 and 0.76 (compared to 0.84 among the 585 non-users), whereas their β_{food} values change much less (0.82/0.87 and 0.89/0.85, against 0.86/0.86). If these individuals were simply more impatient in a domain-general sense, their discount factors would shift uniformly across all rewards. In fact, the questionnaire's direct patience measure moves money and healthy food together: a one-standard-deviation increase raises β_{money} by 0.043 and β_{healthy} by 0.034. In contrast, alcohol and tobacco use are associated with a domain-specific tilt: a lower β_{money} and essentially unchanged food factors. Three additional checks support this interpretation. First, the money-specific gap persists after controlling for subject × reward fixed effects – which absorb baseline impatience and utility levels – as well as for trait self-reports and individual covariates (wealth, BMI, age, gender, trust, and choice consistency). Second, in a reverse regression that includes PB measures from all three domains jointly, only monetary PB significantly predicts drinking (+0.41) and smoking (+0.51). Third, drinkers do not report higher subjective valuations of cash, providing evidence against differential marginal utility of money as the driver of these results.

Figure 10: A trait shifts the profile; drinking and smoking tilt it toward money.



The findings in this section reject both the view of PB as a stable individual trait and its interpretation as an intrinsic property of the commodity class. Rather, PB attaches to the specific (person, reward) pair within a given trade-off structure and state.

7 Magnitude, Welfare, and Policy Implications

7.1 Bounds on β and Behavioral Type Shares

Rejecting the stable-trait specification does not make the magnitude of PB uninformative. Within a given setting, β still summarizes the weight placed on the immediate component for the states, goods, structures and people that the dataset covers. Two questions remain well defined: which values of β can be empirically ruled out, and what proportion of individuals exhibits PB. I address the former through three analytical frameworks of decreasing structural stringency – ranging from parametric estimation to RP sets with no functional form assumptions on utility (Online Appendix F).[33]

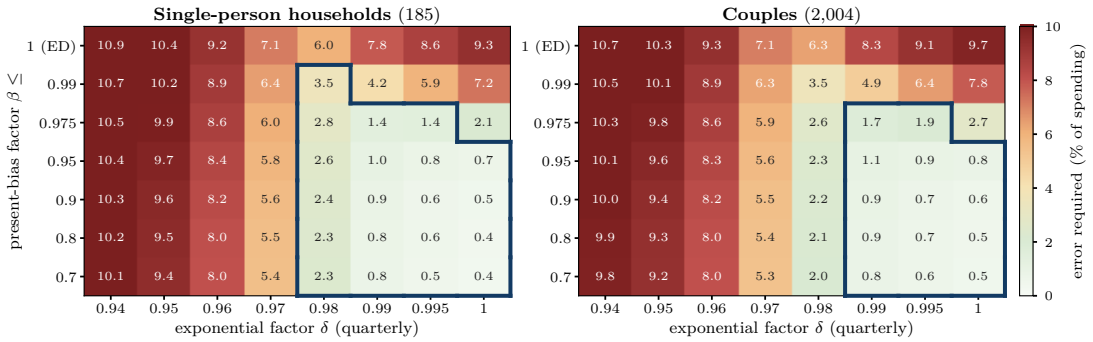
[33]The *parametric* estimate is the two-step estimator of Section 3.2. The *measurement-error-robust* set relies only on the wage-matching identity, a monotone supply curve for each subject, and centered measurement error; it extends the framework of Aguiar and Kashaev (2021) to the quasi-hyperbolic model. The *nonparametric* bracket assumes monotone supply alone and bounds β between wage grid points; by construction, it cannot exclude $\beta = 1$.

In the effort domain, PB is systematically detected: in Augenblick and Rabin (2019), the hypothesis that no wage quintile displays a positive plan-versus-act wedge is rejected ($p = .004$) using the joint moment-inequality test of Cox and Shi (2023). In the same panel, the parametric estimate is 0.80 (the state-weighted average of Section 5.1), the measurement-error-robust (MER) set is $[0.75, 0.95]$, and the nonparametric bracket is $[0.72, 1.00]$. In the independent real-effort study of Fedyk (2024), the corresponding figures are 0.74 $[0.68, 0.80]$, $[0.80, 0.95]$, and $[0.67, 1.00]$. Exponential discounting is rejected at the MER level in both experiments; in AR2019 the identified intervals reflect genuine behavioral dispersion rather than estimation imprecision alone.

The same empirical question applies to household consumption in real-world settings. The Spanish household panel (ECPF) is a standard testing ground for RP over time (e.g., Adams et al., 2014). On these data, the deterministic test of Browning (1989) rejects exponential discounting (ED) for 78% of single-person households and 82% of couples, and the quasi-hyperbolic discounting (QHD) conditions of Blow et al. (2021) rationalize nearly half of those rejected under ED. Aguiar and Kashaev (2021) show that ED rejections disappear once each quarter's expenditure is allowed to carry centered measurement error. Under that error model, however, goods can be freely reallocated within each quarter while preserving observed total spending, rationalizing every household and leaving the test unable to distinguish among discounting models. Instead, centering the error within eight broad groups of goods each quarter (the aggregates of Adams et al., 2014) preserves robustness to measurement error while restoring power. I therefore extend the MER test for ED developed by Aguiar and Kashaev (2021) to the QHD conditions of Blow et al. (2021) (Online Appendix F.2). Figure 11 shows that ED is rejected at every value of δ, whereas QHD with $\beta \leq 0.95$ and δ close to one is not. Specifically, the measurement error needed to rationalize every household is at least 6% of spending in each goods group under ED, compared with about 1% under QHD.[34]

The second question concerns the population prevalence of PB. Using revealed and predicted PB over effort and money from Cerrone et al. (2026), I bound the population share of each behavioral type while accounting for the sampling error of each measure.[35] Between 67% and 79% of subjects are time-consistent, between 9% and 32% are naïve present-biased, and at most 11% are sophisticated (with residual shares bounded at 1.5% for future-biased and 12% for money-driven types). PB is thus empirically present but confined to a minority of subjects, many of whom fail to anticipate it.

Figure 11: The measurement error that (β, δ) pairs require to rationalize the ECPF data.



Note: Each cell reports the average error, in percent of spending in each group of goods and quarter, needed for every household to satisfy the QHD conditions. The blue line encloses the pairs not rejected at the 5% level by the MER test. ED needs errors of at least 6% of spending, QHD with $\beta \leq 0.95$ and $\delta \geq 0.99$ about 1%. Without measurement error, every pair is rejected.

[34] This is not direct evidence of PB; unlike experiments, observational consumption paths bound β only from above.

[35] Each participant is placed in one of eight cells by three binary signals: PB over effort, PB over money, and whether the prediction anticipates the sign of the effort wedge. A mixing matrix gives the probability that each of five types lands in each cell, with noise calibrated to each measure's within-subject sampling error, and a linear program bounds the type shares over a simultaneous 95% confidence box around the observed cell shares (Online Appendix F.3).

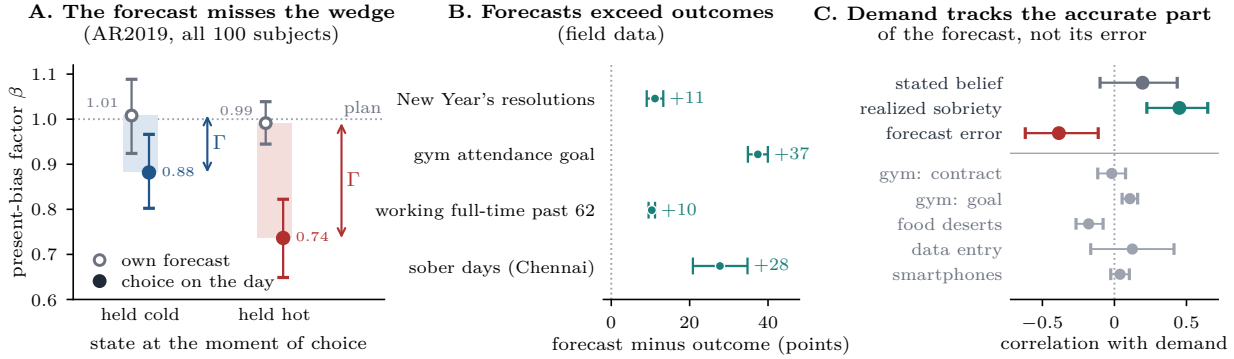
7.2 Welfare Losses, Commitment, and Policy Design

Construing a plan-versus-act wedge as a self-control *problem* inherently requires deciding whose preferences count. The *planning* self and the *acting* self rank the same options differently, and choices alone cannot determine which evaluation should prevail (Bernheim and Rangel, 2009). To limit arbitrary reliance on any single normative standpoint, this section adopts a deliberately conservative criterion. The objective is to establish a welfare benchmark that can secure broad agreement across normative perspectives – accepting a higher detection threshold with the aim of ensuring that the resulting welfare claims remain defensible under different evaluative frameworks.

Specifically, a deviation that an individual did not anticipate is an error in forecasting their own behavior. Choices that rest on this forecast – such as whether to demand commitment or which contract to accept – are made under a misperception of their consequences, and are therefore revealed mistakes rather than evidence of preferences (Bernheim and Rangel, 2009). I therefore define the *self-control gap* Γ as the unanticipated component of the wedge (the shortfall of the realized choice relative to the person's own forecast of it) and treat only Γ as welfare-relevant.

While an anticipated deviation ($e_{\text{plan}} > e_{\text{act}} = e_{\text{pred}}$) reflects PB, it constitutes a conflict between temporal selves on which no welfare claim is asserted. Although this conservative filter may leave some genuine welfare losses undetected, its value lies in establishing a defensible baseline for policy action – one whose mandate for intervention does not depend on the normative position adopted. Crucially, this benchmark is not empty: as shown below, most of the observed wedge is unanticipated.

Figure 12: Most of the wedge is unforeseen, and commitment demand does not track the forecast error.



In AR2019, subjects also make incentivized predictions of the choices they will make on future work days, at five wages drawn at random. Re-estimating them in the two-step coordinate of Section 3.2, separately by assigned state, yields $\beta^{\text{pred}}_{\text{cold}} = 1.01$ and $\beta^{\text{pred}}_{\text{hot}} = 0.99$, against realized values of $\beta^{\text{real}}_{\text{cold}} = 0.88$ and $\beta^{\text{real}}_{\text{hot}} = 0.74$ (Figure 12A).[36] The forecast thus captures virtually none of the realized wedge – not even the additional gap induced by the decision-time state (in the hot state, the prediction anticipates only 3% of the wedge). Field data exhibit the same pattern of misprediction (Figure 12B).[37] Forecasts exceed realized outcomes across diverse domains: by 11 points for New Year's resolutions (Oscarsson et al., 2020), 37 for gym members' stated chance of meeting their own attendance goal (Carrera et al., 2022), 10 for the probability of working full-time

[36] AR2019 report the same near-complete naïveté (a perceived PB factor near 1), and not the split by state.

[37] The gaps, horizon profiles and updating rate in this paragraph are my computations from the replication data; unless stated otherwise, the source papers do not report them. Resolvers' belief in success and their self-rated success are on 0–100 scales. Carrera et al. (2022) report the over-optimism in visits (11.5 forecast, 7.2 realized, without incentives). Schilbach (2019) describes the drivers' beliefs as somewhat overconfident but fairly accurate on average over the whole study; the gap here covers the two weeks before the commitment choice, which the test below requires.

past age 62 in the Health and Retirement Study (HRS), and 28 for the share of sober days that cycle-rickshaw drivers expect under incentives (Schilbach, 2019). These errors are not attenuated by temporal proximity, incentives for accuracy, or experience. In the HRS, the gap does not narrow as individuals approach the target age. Smartphone users underpredict their future screen time almost as much when the forecast period is about to begin as six weeks ahead, and Allcott et al. (2022) find that a larger reward for accurate prediction leaves the error level unchanged. Finally, New Year's resolvers revise their expectations by only a quarter of each month's realized surprise.

Given these welfare concerns, a possible alternative to policy intervention is self-management (Krügel and Uhl, 2023): an individual struggling with time inconsistency can voluntarily commit in advance. Yet voluntary commitment addresses the welfare gap Γ only if those whose predictions are (most) mistaken are also those who seek commitment. The existing evidence is mixed: measured PB is positively associated with commitment demand in ANS2015 and Kaur et al. (2015), negatively in Sadoff et al. (2020), and largely unrelated to it in Carrera et al. (2022), where take-up even declines when awareness is raised. The data of Schilbach (2019) allow a sharper test, because each driver's forecast error can be measured over the two weeks preceding the choice between sobriety incentives and unconditional payments. Schilbach shows that drivers' stated beliefs about their future sobriety predict demand for the incentives; among the 63 drivers whose forecast error is measured before the choice, the correlation is $+0.20$. Decomposing the belief into its accurate part and its error, however, I find that demand correlates positively with the former ($+0.45$ with realized sobriety) but negatively with the latter (-0.39; Figure 12C). Greater overestimation of sobriety is thus associated with lower commitment demand. In four further datasets, the correlation I compute between commitment demand and the individual wedge is small and of both signs: $+0.11$ in Carrera et al. (2022), -0.18 in Sadoff et al. (2020), $+0.12$ in Kaur et al. (2015), and $+0.04$ in Allcott et al. (2022), where demand for limit tools does correlate with *perceived* self-control problems. By contrast, external observers perceive the bias more accurately: Fedyk (2024) shows that people anticipate PB in others but not in themselves. In the two-step coordinate, subjects' actual β over effort is 0.74, their assessment of their own β is 0.97, and their belief about others' β is 0.92: a commitment target set by someone else on the basis of the latter belief would close 23% of the gap between self-assessment and behavior. Taken together, these results indicate that the decision to commit is itself made under the mistaken forecast: those who need commitment most are often the least likely to seek it. Thus, the error that defines Γ simultaneously disables the remedy that would correct it.

Three policy implications follow, independently of any normative ranking across temporal selves. First, voluntary take-up of commitment devices cannot reliably identify who needs them: interpreting take-up as a screening tool risks measuring who is already in control. Analogously, declining commitment is not evidence of a preference for flexibility when it rests on a forecast error. Second, the instruments that reach Γ are those that do not depend on the person's own forecast, such as defaults, targets and schedules set by others, as well as incentives offered to everyone, which Carrera et al. (2022) show to be more efficient than commitment contracts. Third, part of the wedge reflects the decision context rather than a fixed individual trait: the same people deviate further from the same plans after a block of work, while their plans and forecasts remain unchanged (Section 5). Properties of the choice environment – such as the timing, the state, and the options – exert a substantial influence on vulnerability to PB and realized choices. Where employers, schools, and digital platforms shape these settings, leaving them entirely unconstrained is itself a choice of environment rather than a neutral baseline. Structuring this decision context thus constitutes a powerful policy instrument. Ultimately, the case for intervention is not a judgment on the person: it addresses an error that people make about themselves, under conditions that others largely set and that often supply the very cues to which the impulse responds (Section 8).

8 Microfoundations: A Multi-System Control Framework

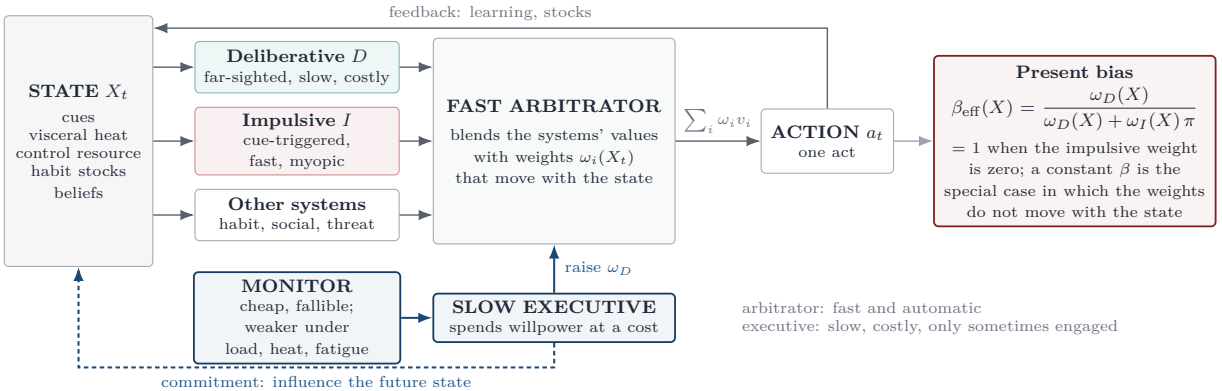
While these findings do not depend on a specific structural model, collectively they motivate a unifying framework. PB shifts with the state, exhibits individual heterogeneity in one trade-off structure but not another, and attaches to the interaction between individual and reward type rather than to either in isolation. This section outlines a framework that accounts for these patterns while nesting standard models as special cases. It is presented primarily as an organizing interpretation of the evidence: the empirical analysis tests specific restrictions implied by the framework rather than the structural model itself. The idea that the weight attached to the present depends on the decision-maker's state dates to Loewenstein (1996), and cue-triggered choice models subsequently formalized it (Laibson, 2001; Bernheim and Rangel, 2004; Loewenstein and O'Donoghue, 2005). In a related spirit, Pennesi (2020) derives an axiomatic quasi-hyperbolic representation in which the PB factor $\beta_\theta = 1 - \lambda(\theta)$ decreases with cue intensity θ. The framework developed below delivers an analogous state-dependent PB factor, while incorporating the additional structure required to accommodate the empirical evidence across reward domains and trade-off structures.

In the *Multi-System Control* (MSC) framework (Figure 13), individual decision-making is governed by several parallel valuation systems, each appraising options over its own horizon. In the spirit of dual-process accounts (e.g., Thaler and Shefrin, 1981; Loewenstein and O'Donoghue, 2005; Kahneman, 2003), these include a forward-looking, slow and costly *deliberative system* D, and a fast, cue-triggered and myopic *impulsive system* I. The state X_t (cues, visceral states, depletable control, habits, beliefs) governs the relative *weight* assigned to each system rather than the fundamental valuations. A fast and automatic *arbitrator* selects the action

$$a_t = \arg\max_a \sum_i \omega_i(X_t)\, v_i(a), \qquad \omega_i(X_t) = \frac{\exp \eta_i(X_t)}{\sum_j \exp \eta_j(X_t)},$$

where $v_i(a)$ is the value that system $i \in \{D, I, \dots\}$ assigns to action a and the weights ω_i depend on the state. An *executive control* process can increase the deliberative weight ω_D by drawing on a finite control resource. It is recruited only when a monitoring system – whose precision degrades under cognitive load, visceral heat, and fatigue – judges that the benefits of control outweigh its costs. Beyond immediate choices, the executive can act dynamically by modifying the state under which a subsequent decision takes place – the core mechanism of commitment.

Figure 13: The MSC — state-dependent weights combine parallel valuations into one action and determine PB.



This architecture is consistent with neuroeconomic evidence: immediate rewards predominantly engage limbic valuation areas (the analogue of I), whereas the lateral prefrontal cortex (closer to D) is engaged by all intertemporal choices (McClure et al., 2004). Crucially, Figner et al. (2010) show that disrupting the left lateral prefrontal cortex raises choices of immediate rewards while leaving choices between two delayed rewards unchanged – mirroring the decision-time state effect identified in Section 5. Whether these neural dynamics reflect active inhibitory control or integrated value comparison remains an active debate (Bulley et al., 2022).

PB follows from the state-dependent weighting of parallel valuation systems. For instance, in an intertemporal trade-off where the impulsive system I evaluates only the immediate payoff, the arbitrator ranks options under an effective PB factor:

$$\beta_{\text{eff}}(X) = \frac{\omega_D(X)}{\omega_D(X) + \omega_I(X)\,\pi} \in (0,1], \qquad \pi \equiv \frac{I\text{'s value of the immediate component}}{D\text{'s value of it}}.$$

When the impulsive weight is zero ($\omega_I(X) = 0$), $\beta_{\text{eff}}(X) = 1$ and decision-making reduces to standard exponential discounting; conversely, as cues, visceral heat, or temporal proximity raise $\omega_I(X)$, $\beta_{\text{eff}}(X)$ decreases. An evolutionary perspective suggests why such a fast, cue-driven system exists: self-control failures can arise from well-learned strategies that are adaptive in the broader settings in which they evolved, for instance where waiting carries a risk of losing the reward (Hayden, 2019). However, they misfire in modern environments that manufacture such cues faster than genes can adapt (Burnham and Phelan, 2020).

Standard models can be represented as special cases of this framework. With a single system and no control cost, the agent discounts exponentially. Under two systems whose weights are invariant to the state, β_{eff} is constant and the framework reduces to quasi-hyperbolic discounting (the special case that Sections 5–6 reject). A single self-control cost, common to all temptations, corresponds to the dual-self model of Fudenberg and Levine (2006), and a habit stock with a cue-triggered impulsive weight to the addiction model of Bernheim and Rangel (2004).[38] Relative to Pennesi (2020), incorporating the executive, the monitor, and the domain-specific allocation of control yields predictions that a scalar cue-indexed weight does not generate: the distortion attaches specifically to the immediate component of a tempting reward, and the PB factor can vary across individuals in one trade-off structure while remaining uniform in another. Finally, the state-dependent factor that the framework implies is characterized over budgets in Section 2.1, so that it has empirical content on the data the literature collects.

The framework further provides an interpretive lens for the monetary findings in Section 6. As a relatively recent innovation in evolutionary terms, liquid cash possesses no intrinsic visceral appeal; PB over monetary payoffs instead reflects what funds can purchase at the time of choice. Where cash implies no immediate consumption – as may be the case for the university subjects in ANS2015 – the model predicts little PB over monetary rewards and minimal interpersonal dispersion. Conversely, where liquidity offers immediate access to visceral goods – as may occur among the adolescent smokers and drinkers in CTW2022 – the MSC architecture is consistent with stronger PB over cash than over food. Findings that conflict under a commodity taxonomy may thus be reconciled within a single state-dependent model.

[38]The MSC framework differs from the dual-self model on two testable margins. First, the single willpower cost of dual-self models implies one β per person, which the evidence of Section 6 rejects. Second, a cue that raises salience without changing resources moves choices under this framework, but leaves them unaffected under a pure willpower model.

9 Conclusions

PB is often treated as a property of the person: a single β that persists within an individual across goods, contexts and moments. This paper has tested that view against the invariances it implies: it fails across states, trade-off structures and goods, while relative individual rankings remain stable within a single structure. Within the same individuals, a randomly assigned state at the moment of choice moves the act and leaves the plan unchanged. Across people, individual PB varies over effort but not over money. In a protocol that varies only the reward, money is the commodity exhibiting the strongest PB – yet this distortion attaches to the person–reward pair rather than to either component in isolation. Uncovering these patterns required a valid testing framework: statistics based on the level of the wedge, conventionally used to document heterogeneity in β, cannot distinguish genuine state dependence from stochastic variation under a state-independent model. In contrast, design-recentered tests of equality across assigned states successfully separate the two. The welfare consequences follow: because the state moves the act while leaving the plan and the forecast unchanged, most of the wedge is unforeseen, and the decision to commit, guided by that forecast, does not track the error. The MSC framework derives this reaction function from state-dependent weights on parallel valuation systems and nests quasi-hyperbolic and dual-self models as restricted cases.

These findings and their limitations point toward actionable avenues for future experiments. The state variation in AR2019 is also consistent with a transient shock to task costs. The main qualitative lesson holds under either interpretation – the aggregate β averages across states – but investigating the underlying mechanism in greater depth requires a purpose-built design. Moreover, the results on goods rest on a single protocol in which money always appears in the same position. Finally, relying on existing datasets naturally restricts the observable range of states and commodities. These considerations motivate three concrete design features. First, factorially combining a randomized consumption cue with dated trade-offs over effort and money would cleanly separate a state-dependent β from a transient cost shock, while testing the MSC framework against a pure willpower model. Second, randomizing the display position of each reward and systematically varying the consumption opportunities afforded by cash at fixed dates would isolate the option value of money from the good itself. Finally, repeated elicitations for each individual within each trade-off structure would distinguish true heterogeneity from measurement imprecision.

Ultimately, these findings reframe how estimated discount factors should be interpreted. A single estimate of β reflects not an immutable trait of the person, but a conditional summary of a state-dependent reaction function, averaged over the support of states, commodities, and trade-off structures sampled in a given dataset. In this view, PB is less a fixed property of who people are than a systematic response to the context in which they choose. This reinterpretation underscores the critical role of targeted policy design. Across domains as diverse as retirement saving, health behaviors, digital engagement, and substance use, present-biased choices can be amplified by an evolutionary mismatch between ancestral decision heuristics and modern environments engineered to capture attention and exploit impulse. A static personal trait invites identification and targeting; a state-dependent reaction function, by contrast, calls for attention to the settings in which decisions are made. Transcending normative debates on whether to fix or penalize the individual, the design of choice environments provides a directly actionable margin for protecting long-run welfare while respecting human agency.

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